

The Entropic Scree: An Information-Theoretic Mapping Framework for Estimating Intrinsic Rank and Informational Gravity in Tabular Systems

Terrence J. Lee-St. John, PhD

TERRY@ENLI.COM.AU

Enli

Abu Dhabi, UAE

Editor: Under Review

Abstract

Linear covariance and distance-based neighborhood geometry have established limitations for representing nonlinear dependence and high-dimensional data. This paper introduces the Entropic Scree, an information-theoretic diagnostic based on a joint-entropy-normalized mutual information matrix for discrete and discretized measurements. Its normalized variation-of-information dissimilarities provide a metric representation; centering and eigen-decomposition supply interpretable informational axes. An equal-axis map of retained eigenvector loadings supports exploratory variable clustering without letting dominant eigenvalues control geometric weighting. A complementary eigenvalue-scaled map relates coordinates to information dissimilarities, with explicit effects of positive-spectrum projection and coordinate omission. Logarithmic spectral decay is used to select a primary rank and an Extended Signal Tail. A spectral-volume decomposition, Average Informational Gravity (AIG), and Factor-Specific Informational Gravity (FSIG) distinguish the overall shared-signal percentage from the shared informational footprint allocated to primary axes. In the main synthetic experiment, the primary boundary selected 20 axes, matching the 20 generating variables, while the evaluated PCA, rank-PCA and RBF Kernel PCA diagnostics did not yield the same selection. The selected rank remained 20 across the three tested binning settings, although gravity estimates changed substantially. Additional root-dependence experiments retained a boundary near the generating count. These results support an information-based alternative for the configurations studied.

Keywords: Information Theory; Principal Component Analysis; Factor Analysis; Sklar's Theorem; Copula Theory; Non-linear Manifold Learning; Dimensionality Reduction; Variable Mapping; Exploratory Variable Clustering

1 Introduction

Dimensionality estimation determines how much structure a downstream representation can express. Too few retained directions can merge distinct patterns, whereas too many can admit unstable detail and increase the demands placed on subsequent models. The relevant dimension depends on the question: the number of coordinates needed for a linear reconstruction can differ from the number of variables in a generating mechanism.

For example, measurements X , X^2 and X^3 are all determined by one underlying variable X , yet they generally span several linear directions. A method can therefore represent their observed variation correctly while giving a dimension that would be misleading if interpreted

as a count of generating mechanisms. This paper develops an informational diagnostic for examining that distinction in mixed tabular data, where continuous measurements, thresholds and categorical states may provide different views of overlapping processes.

Principal Component Analysis (PCA) and inspection of its eigenvalue decay are widely used diagnostics of linear representational dimension (Cattell, 1966). PCA finds orthogonal directions that maximize retained variance, equivalently minimizing squared reconstruction error at a specified linear rank. This descriptive optimization does not require Gaussian data (Jolliffe and Cadima, 2016). Its limitation for the present purpose is substantive: covariance can miss nonlinear dependence, and a low-dimensional generating system can require a much larger linear representation.

Entropic Scree addresses that distinction by replacing covariance with normalized mutual information among measurements. Continuous, ordinal and categorical measurements are represented through discrete probability tables, providing a common dependence scale. The framework is intended to diagnose informational structure in settings where linear rank and generating dimension diverge; the experiments examine this objective in the stated mixed-data systems.

The resulting map has one point per measurement, rather than one point per observation. Its purpose is to show how measurements share information, which informational directions dominate that configuration, and how much spectral weight extends beyond the primary region. Bipolar modules summarize the extremes of individual axes, while the joint coordinates reveal proximity and separation across several informational contrasts. Measurements sharing a pole on one axis can be distinguished on others, allowing exploratory variable groupings to be examined beyond the separate pole summaries. Entropic Scree therefore examines observed variable relationships before any downstream model constructs a representation of individual records.

Entropic Scree develops an integrated framework for interpreting the geometry and spectrum of pairwise information relationships, connecting measurement mapping with dimensionality selection and informational-gravity diagnostics. It builds on normalized variation of information (Rajski, 1961), classical scaling (Torgerson, 1952), entropy effective rank (Roy and Vetterli, 2007), and earlier spectral analyses of information matrices. The framework connects bipolar loading interpretation with joint-coordinate geometry, making explicit how normalization, centering, positive-spectrum projection and coordinate retention affect the relationships being interpreted. The experiments evaluate the proposed spectral-boundary interpretation in the specified synthetic systems.

The value of this integration is that dimensionality selection and variable interpretation can be examined together: a retained spectral region specifies which informational distinctions enter the map, the loading configuration identifies the measurements expressing those distinctions, and the gravity summaries describe the shared spectral volume allocated to the selected primary structure. This last distinction is practically important in noisy systems. A small overall shared-signal percentage can coexist with a substantial shared informational footprint per primary axis, particularly when many measurements express a comparatively compact primary structure. AIG and FSIG make that footprint explicit, on average and for individual axes, respectively.

Information-specific terminology distinguishes the variable relationships represented here from covariance-based interpretations. Informational axes describe contrasts among

measurements' information relationships, and their opposite poles do not denote negative correlation. The terms introduced below pair these interpretations with explicit mathematical definitions: spectral effective rank describes the distribution of spectral weight, volume expresses the measurement-count accounting, and gravity summarizes its allocation across selected informational axes. Table 1 summarizes the principal terms and notation; the following sections provide their definitions and interpretation.

Table 1: Terminology and notation used in the Entropic Scree. Volume quantities use the retained-measurement reference of m variable equivalents. Coordinate geometry and spectral-volume accounting are distinct.

Term	Symbol	Meaning
Prepared observations	N	Number of common prepared records entering empirical probability tables.
Retained measurements	m	Number of variables retained after preparation; the reference total for volume accounting. The simulation uses M for the number generated before preparation.
Intrinsic Generative Rank	r	Number of generating coordinates specified by the generating model.
Realized Intrinsic Root Expansion Rank	K_{rlzd}	Number of active expansion terms in the specified generating construction; distinct from the sampled design matrix rank.
Effective Intrinsic Root Expansion Rank	K_{eff}	Entropy effective rank of the active weighted expansion design's covariance spectrum.
Observed Generative Rank	K_{roots}	Selected primary informational boundary, retaining axes 1 through K_{roots} . Its correspondence with r is assessed separately.
Extended Signal Tail boundary	$K_{extended}$	Inclusive endpoint of the broader region used in volume accounting. The additional tail lies after K_{roots} and through this endpoint.
Normalized mutual information	$\mathcal{M}_{ij}$	Joint-entropy-normalized information, $I(X_i; X_j)/H(X_i, X_j)$.
Normalized Variation of Information	D_{ij}	$1 - \mathcal{M}_{ij}$, on the represented discrete information.
Centered information matrix	$\mathcal{M}_c$	$\mathbf{HMH}$, with $\mathbf{H} = \mathbf{I} - \mathbf{1}\mathbf{1}^T/m$.
Raw and working eigenvalues	λ_i, η_i	Raw eigenvalues of $\mathcal{M}_c$ and working weights $\eta_i = \max(10^{-9}, \lambda_i)$, respectively.
Normalized spectral weights	p_i	η_i/S_ε , where $S_\varepsilon = \sum_j \eta_j$.
Total Unique Probabilistic Volume	R_{eff}	$\exp(-\sum_i p_i \ln p_i)$: effective number of directions in the entire working spectrum.
Selected spectral-weight fraction	P_{sig}	$\sum_{i=1}^{K_{extended}} \eta_i/S_\varepsilon$; distinct from the shared-volume fraction TSV/m .
Redundant Signal Volume	$m - R_{eff}$	Difference between the measurement-count reference and effective spectral number.

Continued on next page

Table 1 continued

Term	Symbol	Meaning
Unique Signal Volume	$P_{sig}R_{eff}$	Effective spectral volume allocated to the selected signal region.
Idiosyncratic Informational Volume	$(1 - P_{sig})R_{eff}$	Remaining effective-volume allocation, including noise or structure outside the selected region.
Total Shared Signal Volume	TSV	$(m - R_{eff}) + P_{sig}R_{eff}$. Its percentage of the reference is $100 TSV/m$.
Average Informational Gravity	AIG	TSV/K_{roots} , in variable equivalents per primary axis.
Factor-Specific Informational Gravity	$FSIG_i$	$TSV\eta_i / \sum_{j=1}^{K_{roots}} \eta_j$. Here “factor-specific” means specific to an informational axis.
Structural Topology Profile	T_i	$FSIG_i/FSIG_1 = \eta_i/\eta_1$ over the primary axes.
Negative Spectral Mass Ratio	NSMR	m_-/m_+ , for raw positive mass $m_+ = \sum_i \max(0, \lambda_i) > 0$ and negative mass $m_- = \sum_{\lambda_i < 0} \lambda_i $.
Informational axis and raw loading	v_{ik}	Measurement i ’s coefficient on informational eigenvector k ; not a correlation with an observation-level component score.
Equal-axis coordinates	$Z_{ik} = v_{ik}$	Coordinates over a stated retained positive axis set, with equal total squared weight per axis.
Eigenvalue-scaled coordinates	$L_{ik} = \sqrt{\lambda_k}v_{ik}$	Coordinates over that same set, retaining eigenvalue weights in squared distances.
Bipolar module and anchor groups	—	Paired positive- and negative-pole summaries of one informational axis; each pole supplies an observed-variable anchor group.
Relative informational estrangement	—	Opposing positions in a specified information contrast; the joint map determines the full geometric relationship.
Dimensional Inflation Index	Δ_K	$K_{PCA} - K_{roots}$, with both dimension-selection rules stated.

1.1 Prior Information-Based Mapping and Spectral Analysis

Information-based eigenanalysis predates the present framework. Early supervised feature extraction included eigendecomposition of an information matrix involving feature pairs and class labels (Bollacker and Ghosh, 1996); that supervised construction has a different objective from the unsupervised pairwise map considered here. Wang and Shen (2006) used eigenvalues of a pairwise MI matrix to construct a multivariate image-registration criterion. In process monitoring, Tong and Shi (2015) replaced a correlation matrix with an MI matrix and used its eigenvectors for feature extraction. Lv et al. (2021) subsequently developed projections of a mutual information matrix (PMIM), combining MI eigenvectors with observation-level projections and monitoring statistics. These studies establish a direct precedent for using information matrices beyond linear covariance.

Information-based mapping and clustering of observed variables also predate the present framework. Benjaminsson et al. (2010) constructed the dissimilarity

$D_{ij} = 1 - I(X_i; X_j)/H(X_i, X_j)$ between voxel time series, used iterative metric multidimensional scaling (MDS) to obtain coordinates, and clustered the resulting representation. Their distance is the same joint-entropy-normalized complement used here. Jiang and Yan (2015) subsequently partitioned process measurements using MI-spectral clustering before fitting multiblock kernel PCA models.

At the observation level, the published methods of scMINER specify the same joint-entropy-normalized information distance between cell expression profiles, followed in its MICA-MDS mode by MDS and k -means clustering (Pan et al., 2025). These precedents show that information-distance mapping is an established approach, while also distinguishing the objects being mapped and the embedding procedures used. The term normalized mutual information alone does not identify a normalization or a spectral operator.

Entropic Scree integrates a centered spectral map of measurements with separate primary and extended boundaries and spectral-volume and gravity accounting. Section 3.3 specifies its classical-scaling interpretation and distinguishes it from applying classical MDS directly to NVI distances. Section 6.4 connects bipolar summaries to joint-space proximity and distinguishes equal-axis from eigenvalue-scaled coordinates, separating coordinate weighting from the subsequent choice of a cluster-membership rule.

Earlier work also raises questions about what an information spectrum represents. Zhao et al. (2017) studied MI eigenvalue and eigenvector behavior, including a prominent leading mode in their random-matrix comparisons, and developed a global interaction summary for time series. Jakobsen (2014) established that ordinary discrete MI matrices are not always positive semidefinite, a limitation also acknowledged by Lv et al. (2021). Sections 3.3 and 5 connect these issues to a centered measurement map: they specify the treatment of negative eigenvalues and show algebraically how a common positive similarity level differs from relative informational contrasts.

The information estimator is a separate choice from the matrix being decomposed. PMIM employs matrix-based Rényi entropy functionals constructed from sample kernel Gram matrices (Sánchez Giraldo et al., 2015; Lv et al., 2021). Contreras-Reyes (2021) instead developed a parametric, continuous-distribution MI-matrix analysis motivated in part by histogram sensitivity. The present implementation uses joint-entropy-normalized plug-in Shannon information on discrete or discretized measurements. Its binning ablation distinguishes stability of a selected boundary from sensitivity of spectral-volume allocations. Taken together, the contribution is a diagnostic interpretation of the variable-level spectrum and map, including primary versus extended structure and gravity accounting, rather than a new general-purpose MI estimator or a claim that eigendecomposing MI is itself new.

1.2 Distinction from Topological Intrinsic Dimensionality

Neighborhood-based intrinsic-dimension estimators, including MLE (Levina and Bickel, 2004) and TWO-NN (Facco et al., 2017), infer dimension from local sample geometry. A fundamental limitation is distance concentration. Let $D_{\min}^{(m)}$ and $D_{\max}^{(m)}$ be the nearest and farthest distances from a query to a fixed finite collection of sampled records in dimension m . Under the distributional concentration conditions of Beyer et al. (1999),

$$\frac{D_{\max}^{(m)} - D_{\min}^{(m)}}{D_{\min}^{(m)}} \xrightarrow[m \rightarrow \infty]{P} 0.$$

In that regime, proximity loses discriminatory meaning: the nearest point is barely distinguishable in relative distance from the farthest. This is a limitation of the available distance information, not merely of search speed. Metric choice affects the severity of this problem (Aggarwal et al., 2001). However, $m > N$ alone is not the theorem’s sufficient condition, and structured high-dimensional workloads can retain meaningful neighborhoods.

Sparse sampling, manifold curvature and nonuniform density introduce further difficulties for neighborhood-based dimension estimates; DANCo and geometry-aware MLE explicitly address such problems (Ceruti et al., 2014; Gomtsyan et al., 2019). Mixed measurements pose an additional representation problem: arbitrary numeric coding of nominal categories imposes distances unsupported by category meaning. Suitable mixed-data dissimilarities are possible, but require explicit choices about coding and relative weighting.

Kernel PCA provides a different nonlinear representation whose behavior depends on the kernel and its parameters.

The comparisons below evaluate PCA, rank-PCA and one specified RBF Kernel PCA construction. Entropic Scree changes the object being mapped: instead of comparing records by their locations in a measurement space, it compares measurements by their pairwise information relationships across records. This avoids having to place categorical codes and continuous magnitudes on a common sample-distance scale. It does not remove the need for sufficient observations to estimate those relationships or for explicit discretization choices.

2 Structural Constraints of Linear Estimators

Deploying standard PCA in mixed-data, high-dimensional environments reveals structural limitations across four dimensions:

- **Mixed-Data Penalty:** A deterministic relation between a continuous variable and a thresholded measurement does not guarantee unit Pearson correlation. For example, if $X \sim N(0, 1)$ and $Y = \mathbf{1}\{X > 0\}$, then Y is completely determined by X , yet $\text{Corr}(X, Y) = \sqrt{2/\pi} \approx 0.798$. The resulting linear association also depends on the threshold and marginal distribution (Cohen, 1983; MacCallum et al., 2002). Thus Pearson correlation does not measure deterministic dependence uniformly across measurement types. This motivates an information-based comparison, while leaving the resolution loss caused by discretization explicit.
- **Non-Linear Blindness:** Zero covariance does not imply independence. For a variable X whose distribution is symmetric about zero, with $\mathbb{E}[X^4] < \infty$ and $\text{Var}(X^2) > 0$, $\text{Cov}(X, X^2) = 0$ despite the nonconstant deterministic relationship. A covariance-based representation therefore cannot infer this dependence from the off-diagonal entry. The marginal variance of X^2 remains in the matrix; it is the dependency, not all signal energy, that is missed.
- **Algebraic Rank Constraint:** For an $N \times m$ centered data matrix, the sample covariance rank is at most $\min(m, N - 1)$. Thus, even if a population covariance has 500 nonzero directions, 100 observations support at most 99 nonzero sample covariance eigenvalues. This is an exact algebraic restriction on that representation, distinct from the statistical question of how many generating mechanisms can be identified.

The restriction follows because the sample covariance is proportional to $\mathbf{X}_c^T \mathbf{X}_c$, and the N rows of the centered matrix sum to zero. Increasing the number of measured columns cannot increase this matrix's rank beyond the number of independent centered rows.

- **Nonlinear Expansion and Linear Dimensional Inflation:** A single generating variable can give rise to multiple linearly independent observed functions. For $X \sim N(0, 1)$ and square-integrable f , the centered function has a probabilists' Hermite expansion converging in L^2 :

$$f(X) - \mathbb{E}[f(X)] = \sum_{k=1}^{\infty} c_k H_k(X), \quad \mathbb{E}[H_i(X)H_j(X)] = i! \mathbf{1}\{i = j\}.$$

This expansion is related to the Gaussian dependence calculus underlying Price's theorem (Price, 1958). If observed measurements span several of these functions, their covariance can have multiple nonzero directions although they share one generating variable. A nonlinear threshold generally requires an infinite expansion, but a finite observed matrix can span only finitely many directions. Linear representational dimension can therefore exceed generating dimension.

Concretely, $H_1(X) = X$, $H_2(X) = X^2 - 1$ and $H_3(X) = X^3 - 3X$ are mutually uncorrelated under a standard-normal distribution, yet each is determined by the same X . If the measured functions span these three directions, three components are appropriate for exact linear representation but should not be mistaken for three independent causes. It is the interpretation of linear dimension as generating dimension that fails in this example.

3 Entropic Scree Methodology

The Entropic Scree changes the dependence measure supplied to the eigensolver. Mutual information compares the joint distribution of two measurements with the distribution that would result if they were independent. It can therefore detect dependence expressed through thresholds, curved relationships or changes in conditional spread, even when covariance is zero.

Copula theory explains the role of marginal transformations. Sklar's theorem separates a joint distribution into its marginal distributions and a copula (Sklar, 1959). For continuous marginals, the probability integral transforms $U = F_X(X)$ and $V = F_Y(Y)$ put both variables on uniform probability scales while retaining their copula. When a copula density exists, its entropy satisfies (Ma and Sun, 2011)

$$I(X; Y) = I(U; V) = -h(c_{X,Y}).$$

The dependence can thus be studied without attaching significance to the original units or marginal shapes. More generally, population mutual information is unchanged by invertible transformations of the individual variables.

For discrete measurements, mutual information is calculated directly from category probabilities,

$$I(X; Y) = \sum_{x,y:p(x,y)>0} p(x,y) \ln \frac{p(x,y)}{p(x)p(y)}.$$

No interpolation between category codes is required. Continuous measurements enter the implementation through discretization, while nominal categories enter as distinct states. Thus both contribute to the same type of probability table, although binning resolution and finite-sample estimation can change the resulting information values.

In empirical calculations, $p(x, y)$ is the observed joint count divided by the number N of complete records, and $p(x)$ and $p(y)$ are its empirical marginals. The resulting quantities are plug-in estimates; the same notation is used for them unless a population example is stated explicitly. All pairs use the same prepared records. Thus an empirical zero MI denotes independence in the empirical table, rather than establishing independence in the population.

The representation remains pairwise. If A and B are independent fair bits and $Y = A \oplus B$, every pair is independent even though the three variables jointly obey a deterministic rule. A pairwise MI matrix cannot distinguish this construction from three independent fair bits. This illustrates a specific blind spot; it does not mean that every multivariable interaction is pairwise invisible. Appendix D distinguishes these cases. Downstream nonlinear analysis of the original records can investigate additional structure, but its recovery is not supplied by the pairwise spectrum alone.

3.1 Published Implementation and Data Preparation

The computational method described here is implemented in the published R package **Entropic.Scree**, version 1.0.1. Its principal function, **Entropic.Scree()**, accepts a **data.table**. The historical implementation used for the simulation is distinguished in Section 8.1.

By default, the package checks for constant and identical columns and, where the numeric design and available sample size permit, multivariate linear collinearity. The automatic bin target is

$$B_{bins} = \max\{2, \lceil cN^{1/3} \rceil\},$$

where **bin_multiplier** supplies c and defaults to one; **num_bins** can specify the target directly. Numeric columns with more than B_{bins} distinct values undergo equal-frequency discretization. Other columns are encoded by their distinct states. Nominal measurements should therefore be supplied as categorical columns when their numeric labels are not intended as an ordered measurement scale.

After discretization, the function removes identical encoded columns and columns with marginal entropy below **low_entropy_thresh**, which defaults to 0.05 nats. These two checks remain active when the initial purge and collinearity options are disabled. Throughout the spectral calculations, m denotes the number of measurements retained after preparation.

For version 1.0.1, complete prepared records and a bin target at least as large as the number of encoded states in every categorical column are required for the count calculation described here. Larger categorical vocabularies require increasing **num_bins**; this also changes the resolution target for continuous measurements. These settings should be recorded with the analysis.

3.2 Normalized Mutual Information and Variation-of-Information Distance

An unnormalized MI matrix has diagonal entries $I(X_i; X_i) = H(X_i)$. A finely resolved measurement can therefore have a larger self-information value than a binary measurement, even when both are informative about the system. The present framework chooses a relative-information scale: each nonconstant measurement has self-similarity one, while a pair’s similarity expresses how much of its joint information is shared. This makes the normalization a substantive measurement convention, not merely a numerical rescaling. Constants with zero entropy require removal or an explicit separate convention.

Normalization matters for the metric property. Let A and B be independent fair bits and $C = (A, B)$. For $d_{avg}(X, Y) = 1 - 2I(X; Y)/(H(X) + H(Y))$, one obtains $d_{avg}(A, B) = 1$ but $d_{avg}(A, C) + d_{avg}(C, B) = 2/3$, violating the triangle inequality. Minimum-entropy normalization gives an even stronger violation: the latter two distances are both zero. This supports choosing a normalization with a valid metric complement.

The framework uses the joint-entropy form of normalized mutual information (NMI), corresponding to an information-theoretic Jaccard similarity:

$$\mathcal{M}_{i,j} = \frac{I(X_i; X_j)}{H(X_i, X_j)} = \frac{I(X_i; X_j)}{H(X_i) + H(X_j) - I(X_i; X_j)}$$

For finite nonzero joint entropy, $0 \leq \mathcal{M}_{i,j} \leq 1$. Zero corresponds to independence of the represented discrete variables. One means that each determines the other almost surely, so the represented information is equivalent even if category labels differ. Intermediate values describe shared information relative to the joint information, rather than the direction of a linear association.

This normalization compares shared information with joint entropy. Its complement, $1 - \mathcal{M}_{i,j}$, is Normalized Variation of Information (NVI) (Rajski, 1961), a metric on discrete information representations with informationally equivalent variables identified. The triangle inequality alone does not establish the Euclidean representability of the distances used below. Section 3.3 therefore distinguishes the centered matrix from its positive-semidefinite approximation.

The resolution penalty can be understood with a discrete example. Let X identify four equally likely states, so that $H(X) = 2$ bits, and let Y report only which of two equally sized groups contains the state. Then $H(Y) = 1$ bit, $I(X; Y) = 1$ bit and $H(X, Y) = 2$ bits, giving $\mathcal{M}_{X,Y} = 1/2$. The dependence is deterministic in the direction $X \rightarrow Y$, but Y does not retain all the distinctions recorded by X .

This differs from Pearson’s linearity constraint: the reduction here records unshared resolution on the chosen information scale. In general, for a nonconstant discrete $Y = f(X)$, $\mathcal{M}_{X,Y} = H(Y)/H(X)$. A bijective relabeling gives one; a genuinely coarser partition can give less than one. The ratio is independent of whether entropy is expressed in bits or nats. For continuous data, the interpretation applies to the discretized measurements used in the calculation.

3.3 Double-Centering and Information Eigendecomposition

The rows and columns of $\mathcal{M}$ refer to measurements. To describe their relative configuration, the framework places the origin at their unweighted centroid. With $\mathbf{H} = \mathbf{I} - \mathbf{1}\mathbf{1}^T/m$, define

$$\mathcal{M}_c = \mathbf{H}\mathcal{M}\mathbf{H}.$$

In elementwise form,

$$(\mathcal{M}_c)_{ij} = \mathcal{M}_{ij} - \bar{\mathcal{M}}_{i.} - \bar{\mathcal{M}}_{.j} + \bar{\mathcal{M}}_{..}$$

Thus the same raw pairwise similarity can occupy a different relative position depending on each measurement's relationships with the rest of the dataset. "Above baseline" and "below baseline" refer to this algebraic reference, not to a statistical null expectation.

The connection to classical multidimensional scaling is explicit. Let $D_{ij} = 1 - \mathcal{M}_{ij}$. Since $\mathbf{H}\mathbf{1} = 0$,

$$\mathcal{M}_c = -\mathbf{H}\mathbf{D}\mathbf{H} = -\frac{1}{2}\mathbf{H}(2\mathbf{D})\mathbf{H}.$$

This is the classical scaling construction for candidate squared distances $2D_{ij}$, equivalently for input distances $\delta_{ij} = \sqrt{2D_{ij}}$ (Torgerson, 1952). When $\mathcal{M}_c$ is positive semidefinite, its eigenvalue-scaled coordinates reproduce those distances; otherwise its positive-spectrum projection gives a Euclidean approximation. By contrast, classical MDS taking D_{ij} itself as the input distance would decompose

$$-\frac{1}{2}\mathbf{H}\mathbf{D}^{\circ 2}\mathbf{H},$$

where $(\mathbf{D}^{\circ 2})_{ij} = D_{ij}^2$ denotes entrywise squaring, not matrix multiplication. The two input-distance choices preserve the ordering of the original dissimilarities but assign different relative separations. For example, NVI values of 0.1 and 0.4 have a distance ratio of four when NVI itself is the target distance, whereas their corresponding distances $\sqrt{2D_{ij}}$ have a ratio of two. The square-root transformation therefore reduces the ratio between larger and smaller positive dissimilarities. This changes the target geometry, not merely its overall scale.

In an exact full-dimensional Euclidean representation, the Entropic Scree construction makes squared coordinate distance proportional to NVI, whereas classical MDS applied directly to NVI makes squared coordinate distance proportional to NVI squared. The resulting centered operators generally have different eigenvalues and eigenvectors, so the choice can affect spectral boundaries, volume and gravity summaries, and variable groupings. Positive-spectrum projection and coordinate truncation introduce further changes to the represented distances. Sections 6.1 and 6.4 introduce equal-axis weighting as a separate subsequent choice: retained eigenvectors are used directly to examine group structure across informational directions without weighting those directions by their eigenvalues.

This construction also explains the relationship to kernel methods: it centers a variable-level similarity matrix and decomposes the resulting representation, rather than requiring an RBF kernel between records (Schölkopf et al., 1998). The information-based similarity may be indefinite; PSD projection supplies the Gram matrix used for the Euclidean map.

Write that projected matrix as $B = LL^T$, where each row L_i is a variable's coordinate vector. Then $B_{ii} = \|L_i\|^2$ is its squared distance from the centroid, and $B_{ij} = \langle L_i, L_j \rangle$ is its

inner product with variable j . For nonzero vectors,

$$\cos \theta_{ij} = \frac{B_{ij}}{\sqrt{B_{ii}B_{jj}}}.$$

A positive inner product gives an acute angle; zero gives orthogonality; a negative value gives an obtuse angle. These are geometric relationships to the centroid, not positive, zero or negative correlations of the original observations. Opposing vectors indicate relative informational estrangement in this map; statistical dependence can remain between the represented measurements. If PSD projection changes the matrix, the angles refer to that approximation.

Centering establishes the reference origin for the variable map. Finite-sample error can remain in the estimated pairwise information; sensitivity to discretization is examined in Appendix B.

Eigendecomposition gives $\mathcal{M}_c = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^T$. Unlike a sample covariance matrix, a pairwise MI matrix is not constructed as the Gram matrix of the N centered observation rows, so the covariance bound $N - 1$ does not generally apply. This representational distinction is shared with earlier MI-matrix approaches (Tong and Shi, 2015; Lv et al., 2021). Centering the variable map imposes rank at most $m - 1$. The available observations still determine how reliably its pairwise probabilities and spectral structure can be estimated.

Positive semidefiniteness requires attention because it is not inherited merely by replacing covariance with information. Jakobsen (2014) gives population counterexamples for the ordinary discrete matrix with entries $I(X_i; X_j)$; symmetric, nonnegative entries alone do not ensure a PSD matrix. Joint-entropy normalization and centering change that matrix, so the spectrum relevant here is that of $\mathcal{M}_c$ itself. A Euclidean Gram matrix has nonnegative eigenvalues: each can contribute a real coordinate scale $\sqrt{\lambda_i}$. If $\mathcal{M}_c$ has negative eigenvalues, the complete matrix cannot be represented as LL^T with real Euclidean coordinates. Setting $\lambda_i^+ = \max(0, \lambda_i)$ gives

$$B = \mathbf{V} \text{diag}(\lambda_i^+) \mathbf{V}^T, \quad L = \mathbf{V} \text{diag}(\sqrt{\lambda_i^+}),$$

the nearest PSD matrix in Frobenius norm (Higham, 1988). The positive eigenvalues and eigenspaces are retained exactly under this operation; stability across different samples is a separate question that depends on estimation error and spectral separation (Davis and Kahan, 1970).

The magnitudes of the retained and discarded parts can be summarized by

$$m_+ = \sum_i \lambda_i^+, \quad m_- = \sum_{\lambda_i < 0} |\lambda_i| = m_+ - \text{Tr}(\mathcal{M}_c), \quad \text{NSMR} = m_-/m_+,$$

when $m_+ > 0$. The Negative Spectral Mass Ratio reports discarded negative spectral mass relative to retained positive spectral mass; multiplying it by 100 expresses the ratio as a percentage. A large value calls attention to the difference between the centered similarity and its Euclidean approximation. This ratio describes spectral mass rather than Shannon-information loss. The actual Frobenius projection error is $\sqrt{\sum_{\lambda_i < 0} \lambda_i^2}$, which uses squared rather than absolute eigenvalues.

The R output labels 100 NSMR as **SCDR**. This diagnostic is calculated from the raw eigenspectrum before applying the numerical floor used for the volume summaries below.

3.4 Spectral Entropy and Total Unique Probabilistic Volume

The eigenspectrum assigns different amounts of spectral weight to different informational directions. The exact positive-spectrum map in Section 3.3 uses $\lambda_i^+ = \max(0, \lambda_i)$. For the numerical spectral summaries, both the published package and the simulation implementation use the working weights

$$\eta_i = \max(\varepsilon, \lambda_i), \quad \varepsilon = 10^{-9}, \quad S_\varepsilon = \sum_{j=1}^m \eta_j,$$

and define

$$p_i = \frac{\eta_i}{S_\varepsilon}, \quad \sum_i p_i = 1, \quad R_{eff} = \exp \left(- \sum_i p_i \ln p_i \right).$$

The small floor permits logarithms throughout the working spectrum. All m weights enter these numerical summaries, including directions assigned the floor. The exact-zero counterpart follows by replacing η_i and S_ε with λ_i^+ and m_+ when $m_+ > 0$.

The floor is negligible only when its aggregate contribution is small relative to the retained positive spectral mass. If $m_+ = 0$, there is no nonzero positive-spectrum map to summarize: flooring alone gives m equal weights and $R_{eff} = m$. Such a result should be identified as a degenerate numerical case rather than interpreted as m resolved informational directions.

Effective rank summarizes the effective number of equally weighted spectral directions (Cover and Thomas, 2006; Roy and Vetterli, 2007). If four directions carry equal weights of 1/4, their effective rank is four. If almost all weight lies in one direction, effective rank approaches one, even when many small weights remain. It therefore describes dispersion, not just the count of nonzero directions or the location of an elbow. Here the weights belong to the centered, variable-level information spectrum. This differs from applying an entropy functional to a sample kernel Gram matrix to estimate information quantities (Sánchez Giraldo et al., 2015): the objects being summarized, their normalization and their interpretation differ even when both calculations use eigenvalues.

This spectral effective rank is termed Total Unique Probabilistic Volume within the framework’s measurement-count accounting. “Total” indicates that the entire working spectrum enters the calculation; “unique” refers to its effective number of spectral directions; and “probabilistic” refers to the normalized spectral weights p_i . “Volume” expresses this effective-number summary in the variable-equivalent accounting used in Section 5, rather than a determinant or geometric hypervolume. The quantity is not a decomposition of multivariate Shannon information into unique and redundant parts. For q positively weighted directions, $1 \leq R_{eff} \leq q$. The numerical floor gives $q = m$; exact zero clipping of a centered spectrum gives $q \leq m - 1$.

The spectral summaries and gravity allocations use the original working weights η_i , independently of the coordinate weighting chosen for visualization or clustering. In particular, the equal-axis map introduced in Section 6.1 does not replace those weights with equal values in the entropy calculation. Neither map normalizes each variable’s coordinate vector to unit length. Such row normalization would replace a map Gram entry by its corresponding cosine similarity where defined, producing a further, angular representation. Axis weighting, row normalization and spectral-volume accounting are therefore distinct choices.

3.5 Logarithmic Scale and Signal Boundary

The scree orders eigenvalues from largest to smallest and examines the positive region above the numerical floor. Its horizontal coordinate counts informational axes, while its vertical coordinate shows their spectral weights. Plotting $E_i = \ln \eta_i$ makes proportional changes visible across a wide dynamic range: a decline from 100 to 50 has the same log-gap as a decline from 2 to 1, although their absolute drops differ greatly.

The diagnostic asks two related questions. Where does the primary, more strongly weighted region end? How far does a broader region plausibly treated as shared structure extend before the remaining tail? These define the proposed primary boundary K_{roots} and Extended Signal Tail boundary $K_{extended}$. A primary break can coexist with a more gradual extended region; the second boundary allows that region to enter signal accounting without treating each of its directions as a separate generating root. Here $K_{extended}$ is the inclusive endpoint; the additional tail beyond the primary region consists of axes $K_{roots} + 1$ through $K_{extended}$ and is empty when the boundaries coincide.

As in the classical scree approach (Cattell, 1966), interpretation begins with the curve and its changes of slope, scale and relative separation. The main contribution here is the information-based spectral construction and its interpretation, with automated boundary recommendations supplied as operational aids. A smooth decline can lack a uniquely resolved boundary. The automatic routine nevertheless returns a positive candidate through its selection or fallback paths. When the curve does not resolve a boundary, that lack of separation should accompany the reported candidate.

The Dual-Diagnostic Ensemble first proposes a macroscopic search region. It then uses successive eigenvalue ratios for the primary recommendation and a local trajectory-departure rule for the extended recommendation. On the standard log-gap path, the first gap is excluded so that a very dominant leading axis does not automatically end the search. Low-rank fallback paths can return one. Appendix A describes the published implementation; Section 8.1 records the historical simulation differences.

Both boundary indices identify the last retained axis. A primary selection of $K_{roots} = 20$ compares the twentieth and twenty-first eigenvalues and retains axes 1 through 20. Automatically, $K_{roots} = K_{log}$ and $K_{extended} = \max(K_{log}, K_{tap})$. In an interactive R session, the package allows users to inspect and revise the recommendations; non-interactive execution uses the automatic values. The companion `Update.Entropic.Scree()` recalculates rank-dependent summaries from an existing result without repeating the pairwise MI calculation or eigendecomposition.

Because centering changes the trace, the mean-eigenvalue reference for this matrix is $\text{Tr}(\mathcal{M}_c)/m$. This provides an operational spectral baseline. The boundary recommendations are heuristic; their calibration is discussed in Appendix A.

4 Five Tiers of Dimensionality

The same dataset can have several different dimensional descriptions. A generating coordinate can appear in multiple transformed measurements, those functions can span several linear directions, and unequal weights can make only part of that span prominent. The following five quantities distinguish these levels:

- **Intrinsic Generative Rank (r):** The number of generating coordinates specified by the model. In a simulation this is known by construction; in an observational application, interpreting it as an identifiable mechanistic count requires additional assumptions.

Sample depth and measurement breadth address different limitations. More observations can improve estimation of the distributions already being measured. Additional informative measurements can distinguish generating configurations that look similar through the original measurements. For example, a common clinical outcome may obscure the distinction between two physiological processes, whereas measurements specific to each process may make it observable. This motivates the measurement-breadth perspective discussed in G2G (Lee-St. John et al., 2026).

- **Realized Intrinsic Root Expansion Rank (K_{rlzd}):** The number of active expansion terms in the specified generating construction. For example, one generating variable may supply X , X^2 and X^3 as three active terms. This term count is distinct from the rank of the sampled design: functional dependence, coefficient mixing and the finite-sample ceiling can reduce that rank. It equals a population linear dimension only under the required linear-independence conditions.
- **Effective Intrinsic Root Expansion Rank (K_{eff}):** The entropy effective rank of the covariance spectrum of the active, weighted expansion design. Many terms can be active while relatively few directions carry most of its weight. This quantity describes the generating-term representation, with its standardization and weighting reported explicitly; it is not calculated from the observed NMI spectrum.
- **Observed Generative Rank (K_{roots}):** Operationally, the selected primary informational rank: the number of axes through the primary boundary of the empirical information spectrum. The generative designation expresses its proposed use as a diagnostic of generating structure represented in pairwise information relationships. Its correspondence with the generating count r is evaluated in the simulations.
- **Total Unique Probabilistic Volume (R_{eff}):** The effective number of informational directions under the working-spectrum convention defined in Section 3.4, evaluated across the entire working spectrum. It measures how broadly spectral weight is distributed, including weight outside the primary region. For example, one dominant direction with many very small positive directions can have an effective rank close to one even if a practitioner selects several axes. Thus R_{eff} is not required to exceed K_{roots} or $K_{extended}$.

The empirical spectrum can contain a strongly weighted primary region followed by a more gradual region before the trailing bulk. The framework distinguishes these operationally:

1. **Primary boundary (K_{roots}):** The selected endpoint of the leading region used to guide downstream representation size.
2. **Extended Signal Tail boundary ($K_{extended}$):** The selected endpoint of the broader region treated as signal for the volume calculation. When $K_{extended} > K_{roots}$, the

intervening directions contribute to that allocation without each being interpreted as an additional root.

One motivating hypothesis is that combinations and transformations of shared generating variables leave additional distinctions in their pairwise information profiles. Some of those distinctions can lie outside the small set of primary axes, so a broader spectral region may carry useful structure. Sampling, heterogeneous dependence strengths, discretization and noise can also influence the intervening region.

Including the region through $K_{extended}$ is therefore an explicit accounting choice: it avoids equating “outside the primary rank” with “irrelevant to the system.” The choice retains a potential contribution from weaker shared patterns.

5 Quantifying Informational Gravity (AIG and FSIG)

Informational gravity denotes the allocation of the framework’s shared spectral volume across selected primary informational axes. Its scale is set by the m retained measurements: one variable equivalent is one unit of this measurement-count reference, rather than an identified original measurement. The choice of m keeps the accounting on that scale even though centering changes the trace and removes the common coordinate direction. Using the working spectrum from Section 3.4, define the fraction of spectral weight through the extended boundary as

$$P_{sig} = \frac{\sum_{i=1}^{K_{extended}} \eta_i}{S_\varepsilon}.$$

The framework then defines three allocations that together partition the m -unit measurement reference. Here “signal” refers to the region included by the selected spectral boundary. The allocations are:

1. **Redundant Signal Volume** ($m - R_{eff}$): The difference between the measurement-count reference and the effective spectral number. Greater concentration in fewer directions increases this allocation. It is called redundant volume within this accounting convention; it is not a separately estimated multivariate redundancy functional.
2. **Unique Signal Volume** ($P_{sig}R_{eff}$): The fraction of effective spectral volume assigned to the selected region through $K_{extended}$. Extending the boundary includes weaker directions without increasing the selected primary root count.
3. **Idiosyncratic Informational Volume** ($(1 - P_{sig})R_{eff}$): The remaining allocation. It may include measurement error and estimation effects as well as structure not resolved or retained by the diagnostic. This calculation does not separate those sources.

This accounting is chosen to combine two complementary spectral properties on a common measurement-count scale: concentration into fewer effective directions and coverage by the selected signal region. The difference $m - R_{eff}$ records the concentration-based allocation, while the remaining effective volume R_{eff} is divided in proportion to the spectral weights inside and outside that region. This proportional split gives a linear allocation without introducing an additional regional weighting parameter. For a fixed spectrum, extending the signal boundary transfers volume from the idiosyncratic allocation to the unique signal

allocation while leaving the redundant allocation unchanged. The three quantities sum to m by construction. Their first two terms define Total Shared Signal Volume,

$$TSV = (m - R_{eff}) + P_{sig}R_{eff} = m - R_{eff}(1 - P_{sig}).$$

For example, if $m = 100$, $R_{eff} = 80$ and $P_{sig} = 0.25$, the three allocations are 20, 20 and 60 variable equivalents. Total shared signal is 40, comprising the first two terms; it is not an additional quantity to add to the same 100-unit total.

The centering contribution and the selected boundary both enter this accounting. Under exact zero clipping, independent measurements with population $\mathcal{M} = \mathbf{I}$ give $m - 1$ equal positive eigenvalues and $R_{eff} = m - 1$. If primary and extended boundaries k and e are assigned, with $1 \leq k \leq e \leq m - 1$, then

$$P_{sig} = \frac{e}{m - 1}, \quad TSV = e + 1, \quad \text{AIG} = \frac{e + 1}{k}.$$

One unit arises from centering and e units from assigning part of the flat spectrum to the selected region. A positive allocation therefore does not by itself establish departure from independence; a flat spectrum also supplies no uniquely resolved elbow. A positive numerical floor slightly changes these exact values.

A related example connects spectral shape to the prominent leading modes examined in earlier MI-matrix studies (Zhao et al., 2017). Suppose every off-diagonal normalized similarity equals a , where $0 \leq a < 1$. Then

$$\mathcal{M} = (1 - a)\mathbf{I} + a\mathbf{1}\mathbf{1}^T.$$

The uniform direction $\mathbf{1}/\sqrt{m}$ has eigenvalue $1 + (m - 1)a$, while each of the $m - 1$ orthogonal contrasts has eigenvalue $1 - a$. A large leading eigenvalue can therefore arise from a common positive similarity level without distinguishing any subgroup of measurements. Centering separates that common direction from the relative configuration:

$$\mathcal{M}_c = (1 - a)\mathbf{H}.$$

This example explains why an uncentered global mode and a centered informational contrast answer different questions. Under exact zero clipping, changing a rescales all nonzero centered eigenvalues without changing their normalized weights. For fixed k and e , the allocations above consequently remain unchanged. Gravity summarizes concentration and allocation within the selected spectral representation; pairwise NMI values retain complementary information about the absolute level of normalized dependence. Removing the uniform direction is a geometric reference choice, not a statistical correction for heterogeneous estimation bias.

5.1 Average Informational Gravity (AIG)

Average Informational Gravity summarizes the total allocated shared volume per retained primary axis:

$$\text{AIG} = \frac{TSV}{K_{roots}}.$$

The numerator includes the selected extended region; the denominator counts only the primary axes. In this bookkeeping sense, the method “rebundles” the broader signal allocation into an average footprint for the primary representation. For the 100-measurement example above, $TSV = 40$ and $K_{roots} = 10$ would give $AIG = 4$ variable equivalents per primary axis. FSIG describes the allocation to individual informational axes below.

An algebraic lower bound can be stated explicitly. Let $k = K_{roots}$, $e = K_{extended}$ and q be the number of positively weighted directions under the chosen spectral convention, with $1 \leq k \leq e \leq q$. Ordered normalized weights give $P_{sig} \geq e/q$, while $R_{eff} \leq q$. Consequently,

$$TSV = m - R_{eff}(1 - P_{sig}) \geq m - q + e, \quad AIG \geq \frac{m - q + e}{k}.$$

For exact zero clipping of the centered spectrum, $q \leq m - 1$, yielding $AIG \geq 1 + 1/k$. If a positive floor is included in all m weights, the general argument instead guarantees $AIG \geq e/k \geq 1$. These bounds follow directly from the allocation definitions and the ordered spectrum.

5.2 Factor-Specific Informational Gravity (FSIG)

AIG gives the average allocation, while differences among the primary eigenvalues determine how that allocation is distributed. Factor-Specific Informational Gravity (FSIG) is the corresponding axis-specific allocation; “factor-specific” refers here to an informational axis, not an extracted latent generating variable. It allocates TSV in proportion to the weight carried by each primary informational axis:

$$FSIG_i = \left(\frac{\eta_i}{\sum_{j=1}^{K_{roots}} \eta_j} \right) TSV.$$

For example, an axis carrying 20% of the primary spectral weight receives 20% of TSV. If TSV is 40 variable equivalents, its FSIG is 8. Thus FSIG combines a global quantity (the total allocation) with a relative quantity (the axis’s share of the primary spectrum). Its conservation and average are exact:

$$\sum_{i=1}^{K_{roots}} FSIG_i = TSV, \quad \frac{1}{K_{roots}} \sum_{i=1}^{K_{roots}} FSIG_i = AIG.$$

This allocation is a useful way to compare informational axes within the chosen representation. It is not an independently estimated contribution of each generating root: one generating process may affect several axes, and one axis may summarize relationships among several processes. Unequal representation in the primary and extended regions can therefore complicate a mechanistic reading of FSIG.

If $K_{roots} = K_{extended}$, the selected signal region ends at the primary boundary, and the same allocation formulas apply. This can reflect a compact spectrum, insufficiently resolved weaker structure, or the boundary-selection rule.

5.3 Practical Interpretation of Informational Gravity

Informational gravity describes the strength of the selected primary structure in terms of its allocated shared spectral volume in the data as observed, including the measurement error

and structural uncertainty already present. AIG expresses the average shared informational footprint per primary axis, while FSIG expresses the allocation to each individual axis. These quantities complement the overall shared-signal percentage by distinguishing the proportion of the measurement reference allocated to shared structure from the size of that allocation per primary axis.

This distinction follows directly from

$$\text{AIG} = \frac{TSV}{m} \frac{m}{K_{roots}}.$$

The first factor is the overall shared-signal fraction; the second is the number of retained measurements per selected primary axis. A small shared fraction can therefore coexist with a substantial average shared footprint when the measurement set is large relative to the selected primary rank. Consequently, substantial shared allocations to the primary structure can coexist with dominance of idiosyncratic informational volume.

FSIG further distinguishes whether this allocation is concentrated in a few primary axes or distributed more evenly among them. The practical interpretation combines these gravity values with the separation of the selected spectral region and the measurements expressing its informational contrasts. In this way, the framework describes both the prominence of shared structure within the overall measurement reference and its allocated strength within the selected primary representation.

5.4 Structural Topology Profile

The Structural Topology Profile is a relative spectral-weight profile: it compares each primary-axis gravity allocation with the leading one. Its role is to summarize the balance among informational contrasts for interpretation alongside the variable map:

$$T_i = \frac{FSIG_i}{FSIG_1} = \frac{\eta_i}{\eta_1}, \quad 1 \leq i \leq K_{roots}.$$

The common TSV multiplier cancels. The profile therefore answers a relative question: is the primary representation dominated by one direction, divided among a few strong directions, or distributed more evenly? Four shapes are especially useful to distinguish:

- **Steep initial drop:** One axis carries substantially more weight than the others. A broadly expressed pattern or a major contrast between large groups can produce such a configuration. A centralized shared process is one hypothesis to examine in the loadings; the profile does not itself identify a hub or demonstrate synergy.
- **Nearly flat profile:** The retained axes carry similar weights. A balanced modular organization is one possible explanation, but equal weights do not establish independent roots. Closely spaced eigenvalues also permit substantial changes of axis orientation under small perturbations, even if their combined subspace is stable.
- **Dominant axis followed by a plateau:** A pronounced leading direction coexists with a more evenly weighted remainder. In the proposed combinatorial interpretation, a broad shared pattern may coexist with balanced secondary distinctions. Assessing that interpretation requires examining which measurements define the leading axis and which distinctions remain afterwards.

- **Several leading axes followed by a drop:** Most retained weight is concentrated in a few directions. Multiple large informational domains are a possible interpretation, but their number and membership require the loading configuration rather than the profile alone.

Eigenvalues describe the relative weights of informational directions, while eigenvectors locate the measurements within them. The same eigenvalues can accompany different variable configurations. Interpreting a topology profile therefore combines its shape with the loadings, direct information relationships and substantive knowledge.

6 Distinction Between Diagnostic Mapping and Feature Extraction

In standard PCA, eigenvectors act as linear coefficients for the original measurement coordinates after the specified centering and scaling. Multiplying that prepared data matrix by the retained eigenvectors produces component scores for observations. In Entropic Scree, the eigensolver instead receives relationships among variable distributions. Its coordinates locate measurements in an information map; they are not derived as an observation-level reconstruction rule.

For a suitably numeric data matrix, $\mathbf{XV}$ is algebraically possible; PMIM uses this type of observation-level projection for fault detection (Lv et al., 2021). Its interpretation requires the specified coding and scaling of $\mathbf{X}$. If $\mathbf{X}$ has sample covariance Σ , then

$$\text{Cov}_{\text{sample}}(\mathbf{XV}) = \mathbf{V}^T \Sigma \mathbf{V}.$$

Diagonalizing an MI matrix does not generally diagonalize Σ , so the resulting observation scores can remain correlated even when the MI eigenvectors are orthogonal. Moreover, changing category codes or measurement units can alter those scores without changing the underlying information relationships. Such a product is a separately defined feature transformation, whereas Entropic Scree maps measurements using rows of retained eigenvectors, either with equal axis weight or scaled by the square roots of their positive eigenvalues. MI does not encode whether two measurements increase or decrease together, so the signs of those map coordinates do not supply directional coefficients for latent record scoring.

The map can serve as a diagnostic architectural blueprint: its selected primary rank proposes a representation size, its loading patterns describe observed informational groupings, and its gravity summaries describe spectral allocation. A nonlinear autoencoder or another downstream representation can then operate on the records themselves. This is a proposed use of the diagnostic, not a guarantee that a bottleneck of that size recovers the true causes or avoids overfitting; reconstruction, stability and substantive interpretation remain separate assessments.

6.1 Interpreting Informational Loadings: Bipolar Modules and the Configurational Space

Informational axes locate measurements in the mapped configuration. The raw informational loading is the unit-norm eigenvector coefficient v_{ik} . For a nonempty retained set $\mathcal{K}$ of positive eigendirections of the raw centered matrix, define equal-axis coordinates $Z_{ik} = v_{ik}$ and

eigenvalue-scaled coordinates $L_{ik} = \sqrt{\lambda_k} v_{ik}$, for $k \in \mathcal{K}$. Their Gram matrices are

$$(P_{\mathcal{K}})_{ij} = \sum_{k \in \mathcal{K}} Z_{ik} Z_{jk}, \quad (B_{\mathcal{K}})_{ij} = \sum_{k \in \mathcal{K}} L_{ik} L_{jk} = \sum_{k \in \mathcal{K}} \lambda_k v_{ik} v_{jk}.$$

The first gives each retained axis equal overall weight; the second preserves its eigenvalue weight. On any fixed positive axis, the scaling preserves loading signs, ordering and magnitude ratios. The following polarity interpretation therefore applies to either coordinate convention; the inner-product formulas below use L for the spectrally weighted case.

This gives a practical reading of the two poles:

- **Large same-sign loadings:** The variables jointly make a positive contribution to the represented inner product on that axis. They occupy the same side of this informational contrast and are candidates for an axis-specific cluster.
- **Large opposite-sign loadings:** The variables make a negative contribution $L_{ik} L_{jk}$ to the inner product on that axis, describing relative estrangement along this contrast. Their squared separation along the axis is $(L_{ik} - L_{jk})^2 = L_{ik}^2 + L_{jk}^2 - 2L_{ik} L_{jk}$; opposite signs make the cross-term contribution positive.
- **Loadings near zero:** The variable participates weakly in this particular contrast. It may still participate strongly on other axes; a small loading here does not mean that the variable is uninformative overall.

Other axes can reinforce or offset these inner-product contributions. For example, coordinate vectors $(1, 2)$ and $(-1, 2)$ occupy opposite poles of the first axis but have total inner product $-1 + 4 = 3$. Full-space orthogonality requires the summed inner product to be zero; it cannot be inferred from pole membership alone. Squared coordinate separations behave differently: each axis contributes the nonnegative quantity $(L_{ik} - L_{jk})^2$ to the pair's total squared distance. Additional axes can reveal separation hidden by a single-axis view, but cannot cancel a separation already represented. Section 6.4 develops this joint-distance interpretation.

The sign interpretation differs from signed linear association. In a correlation-based representation, signs can describe the direction of linear covariation, although cross-loadings and imperfect associations are allowed. In the information map, reversing a numeric measurement or bijectively relabeling a category does not remove its mutual information with other variables. Opposing poles therefore do not mean that one measurement decreases when the other increases. They describe contrasting informational positions, not necessarily zero MI, independent mechanisms or opposing causal effects. The overall sign of each axis is arbitrary. Within an exactly repeated eigenspace, orthogonal rotations also preserve the reconstructed map, so its individual axes and pole memberships are not uniquely determined. In that case, interpretation should distinguish a chosen axis representation from relationships preserved by the combined subspace.

Selecting large positive and negative loadings can yield up to two descriptive anchor groups per retained axis. Across K_{roots} axes this gives up to $2K_{roots}$ pole descriptions, not a census of that many unique clusters or generating roots. A variable or group may recur across axes because several different contrasts involve it. Here a bipolar module denotes the

paired positive- and negative-pole summaries of one informational axis; each individual pole supplies an anchor group. These descriptive groups can overlap across axes and are distinct from a joint-space partition of the measurements.

When `extract_bipolar_modules=TRUE`, package version 1.0.1 returns the eigenvector coefficients v_{ik} and their variable names. By default, `bipolar_top_n=0.20` retains up to $\max\{1, \lfloor 0.20m \rfloor\}$ coefficients separately from each sign, ordered by absolute magnitude. The count is based on all m retained measurements, with fewer returned if a pole contains fewer members. Users may request another proportion or an absolute count. For a retained positive axis, these raw coefficients are the equal-axis coordinates Z_{ik} . Multiplication by $\sqrt{\lambda_k}$ gives the alternative coordinates L_{ik} and preserves within-axis ordering and signs.

For exploratory variable grouping, the proposed primary representation is the equal-axis map Z . Once informational directions have been retained for interpretation, each receives the same total squared coordinate weight, so distinctions along a weaker retained axis are not compressed solely because its eigenvalue is smaller. For example, if two retained eigenvalues differ by a factor of 100, equal differences in their raw loadings contribute 100 times as much squared separation on the stronger axis in L , but equally in Z . Equal-axis weighting can therefore make secondary group distinctions easier to examine alongside a dominant contrast.

The complementary map L preserves spectral weighting and supplies the information-distance identities in Section 6.4. Both maps use the same retained eigenvectors and include every retained measurement, rather than only the displayed pole anchors. Their coordinates, distances and any subsequent clustering should use a stated convention and axis range. The geometric definitions use positive raw eigenvalues, not the numerical floor employed for spectral-volume accounting. Choosing a map changes how the retained configuration is compared; it does not alter the eigendecomposition, boundary selections or gravity summaries.

The substantive meanings of the anchors depend on what the dataset measures. In a largely modular system, an anchor may mainly contain measurements associated with one process. In an entangled system, it may contain measurements jointly influenced by several processes—for example, clinical events involving both cardiac and respiratory function. Such combination-defined groups can be prominent when they generate distinctive dependence profiles. Pure-process measurements can occupy intermediate or different positions, but there is no general rule placing them on shorter vectors or relegating them to trailing axes.

The Structural Topology Profile helps describe how strongly these contrasts are weighted; it cannot choose between the organizational interpretations by itself. Interpretation should inspect the pole-defining measurements, their full coordinate relationships and their substantive meanings. The poles map the observed combinations represented by the data, while extraction of latent generating processes remains a separate task.

6.2 Informational Boundaries and Generating Dimension

A system with r generating coordinates can produce many observed measurements and combinations without introducing additional independent generating inputs. This motivates a central question: can an information-based representation reveal the underlying generating count despite the diversity of its observed expressions? Entropic Scree investigates this

possibility by examining how shared information is organized across the spectrum of the normalized mutual information matrix.

The proposed rationale is that measurements expressing overlapping generating influences can have similar information relationships with other measurements. Such overlap can concentrate their representation into a primary spectral region, even when the observed measurements include numerous nonlinear combinations. Differences in the information retained by individual measurements, together with estimation effects, can require additional directions. This provides a conceptual motivation for distinguishing a primary region from an extended spectral tail, rather than interpreting every retained direction as a separate generating mechanism.

The Data Processing Inequality supports the principle that processing cannot increase information about a source beyond that available in its input (Cover and Thomas, 2006). For the variable map, the question is how the measurements' information relationships are distributed across spectral directions. That distribution depends on the measurement functions, dependencies among generating inputs, discretization and noise.

An exact limiting example clarifies this distinction. Suppose $r \geq 2$ independent, nonconstant, finite-valued roots are each observed through one or more exact copies. Their NMI matrix contains one all-ones block per root and zeros between blocks. Its uncentered rank is r , but its centered rank is $r - 1$. Collapsing exact duplicates gives the same result:

$$\mathcal{M} = \mathbf{I}_r, \quad \mathbf{H} = \mathbf{I}_r - \frac{1}{r}\mathbf{1}\mathbf{1}^T, \quad \text{rank}(\mathbf{H}\mathcal{M}\mathbf{H}) = r - 1.$$

Geometrically, two independent measurement groups require only one bipolar contrast to distinguish their positions in the centered variable map. The number of map directions can therefore differ from the number of generating roots. The $r - 1$ result is specific to this block construction.

Entropic Scree therefore treats K_{roots} as an operational estimate of the primary spectral boundary whose correspondence with generating count is evaluated in the simulations. The observed agreement supports that correspondence in the configurations studied, while a general recovery guarantee remains a separate theoretical question. The paired poles describe extremes of the retained variable map.

6.3 Topological Information Space: An Operational Analogy

Here, “topology” refers descriptively to the organization of measurements and their informational relationships within the map. An operational analogy illustrates the distinction between the original underlying generative space and the vectors the eigensolver actually manipulates.

Consider a system with four independent biological generating processes: Cardiovascular, Respiratory, Neurological and Muscular. Suppose 1,000 clinical measurements include a few comparatively specific indicators and many events jointly influenced by either the Cardiovascular – Respiratory pair or the Neurological – Muscular pair. These give two candidate observed macro-states: Cardiopulmonary and Neuromuscular. Measurements that retain overlapping distinctions about a macro-state can have similar information profiles.

The map locates those measurements by their relationships across records, not by their clinical units. Its coordinates can reveal a contrast between the two macro-states if the

empirical dependence supports it. A limiting example makes the geometry transparent: if every measurement within each group is an exact copy of its own composite state, and the two composite states are independent, the NMI matrix consists of two unit blocks with zero cross-block similarity. Centering produces a one-dimensional contrast, with the groups at opposite poles. Their opposite signs represent separation of independent observed states, not one state increasing as the other decreases.

Less redundant measurements, process-specific indicators, shared influences or measurement error can produce a more complex matrix structure. Several axes can be needed to represent the observed relationships, and the pole assignments depend on the whole configuration. The resulting vector lengths describe the measurements' distances from the centroid under the chosen coordinate weighting; equal-axis and eigenvalue-scaled lengths answer differently weighted questions about that configuration.

The analogy separates three objects. The generating model specifies four processes; the observed data may emphasize a smaller or more complicated set of clinical combinations; and the positive information spectrum measures how those observed distinctions are distributed across directions. K_{roots} selects a primary region, $K_{extended}$ selects a broader region for accounting, and R_{eff} summarizes the dispersion of the working spectrum defined in Section 3.4. This is why the observed bipolar clusters can be informative without being identified with isolated biological roots.

6.4 Multiaxis Geometry and Exploratory Variable Clustering

Information-distance mapping and MI-based spectral grouping have previously been used to organize observed measurements (Benjaminsson et al., 2010; Jiang and Yan, 2015). The present section develops the relationship between informational coordinates, bipolar summaries and joint-space proximity. A pole identifies measurements participating strongly in one informational contrast. A multiaxis map locates each measurement by its complete coordinate vector over the stated retained axes. Variables that overlap in one projection may separate in another, and measurements can form a compact group in the joint space without occupying extreme positions on any single axis. Coordinate weighting determines which differences contribute most strongly to that proximity.

6.4.1 EQUAL-AXIS GEOMETRY FOR EXPLORATORY GROUPING

For the positive retained set $\mathcal{K}$ defined in Section 6.1, let $q = |\mathcal{K}|$, $Z = \mathbf{V}_{\mathcal{K}}$ and $L = Z\mathbf{\Lambda}_{\mathcal{K}}^{1/2}$. The two squared distances are

$$d_{ij,eq}^2 = \|Z_i - Z_j\|^2 = \sum_{k \in \mathcal{K}} (v_{ik} - v_{jk})^2, \quad d_{ij,sc}^2 = \|L_i - L_j\|^2 = \sum_{k \in \mathcal{K}} \lambda_k (v_{ik} - v_{jk})^2. \quad (1)$$

Because the columns of Z are centered and orthonormal, each has total squared coordinate weight one, whereas the corresponding column of L has weight λ_k . Equal-axis proximity therefore describes similarity across the retained informational directions with equal overall weight per direction. Spectrally weighted proximity instead gives larger-eigenvalue directions greater influence. The rescaling is invertible on this positive retained subspace: it changes distances and relative geometric emphasis without adding information or changing which pairs have identical coordinates.

Algebraically, $Z = L\mathbf{\Lambda}_{\mathcal{K}}^{-1/2}$ removes the relative spectral scales from the retained coordinates. This is a whitening of the measurement-coordinate configuration up to a common normalization constant, not a transformation of the original observation records. The use of eigenvector rows as clustering coordinates has established precedent in spectral clustering (von Luxburg, 2007); the present construction uses selected eigendirections of the centered normalized-information matrix rather than a graph Laplacian. Here it provides a grouping representation in which smaller retained spectral weight does not, by itself, make a distinction geometrically negligible.

The equal-axis Gram matrix is the orthogonal projector

$$P_{\mathcal{K}} = ZZ^T = \mathbf{V}_{\mathcal{K}}\mathbf{V}_{\mathcal{K}}^T, \quad d_{ij,\text{eq}}^2 = (P_{\mathcal{K}})_{ii} + (P_{\mathcal{K}})_{jj} - 2(P_{\mathcal{K}})_{ij}.$$

For any orthogonal $q \times q$ matrix Q , replacing Z by ZQ leaves this projector and every pairwise distance unchanged. Equal-axis joint geometry is therefore a property of the retained subspace, even when individual axis labels and poles change with its basis. This is useful when a coherent group participates in several contrasts rather than occupying one isolated pole.

Axis retention remains essential. Equal weighting gives retained weak directions as much total geometric weight as strong ones; it is useful when those directions express meaningful distinctions, but can also emphasize estimation effects. The primary boundary supplies one proposed range, while the Extended Signal Tail supplies a broader range to examine separately. If a boundary cuts through an exactly repeated eigenspace, the retained subspace itself is not uniquely determined. More generally, its sensitivity depends on perturbations and spectral separation at the retention boundary (Davis and Kahan, 1970).

A complete centered basis illustrates why retaining every direction is not the objective. If Z contains all $m - 1$ orthonormal directions perpendicular to $\mathbf{1}$, then

$$ZZ^T = \mathbf{H}, \quad d_{ij,\text{eq}}^2 = 2 \quad (i \neq j). \quad (2)$$

Every pair is then equally separated. The discriminatory value of equal-axis mapping comes from the selected informational subspace, rather than from accumulating all directions. This complete-basis limit is different from retaining all positive directions when they span a smaller subspace.

An orthogonal two- or three-dimensional display can hide separation under either convention. When $q > 3$, an equal-axis configuration has q equal nonzero singular values, so a best rank-three linear projection is not unique. An SVD overview remains a valid projection, but is not a uniquely preferred view of the groups. Original-axis views, alternative orientations and full retained-space distances complement such displays. Visual separation motivates exploration; membership recovery and stability require evaluation against the intended grouping objective.

6.4.2 INFORMATION PROXIMITY IN THE EIGENVALUE-SCALED SPACE

Suppose first that $\mathcal{M}_c = \mathbf{H}\mathbf{M}\mathbf{H}$ is positive semidefinite and every positive coordinate is retained. Since $\mathcal{M}_{ii} = 1$, the coordinates L defined in Section 3.3 satisfy

$$d_{ij}^2 = \sum_{k:\lambda_k>0} (L_{ik} - L_{jk})^2 = 2(1 - \mathcal{M}_{ij}). \quad (3)$$

To see this directly, let e_i be the i th coordinate basis vector. The contrast $e_i - e_j$ is unchanged by $\mathbf{H}$, so

$$(e_i - e_j)^T \mathcal{M}_c(e_i - e_j) = (e_i - e_j)^T \mathcal{M}(e_i - e_j) = 2(1 - \mathcal{M}_{ij}).$$

Thus greater normalized mutual information corresponds exactly to smaller full-space distance in this case. It is the pairwise separation, rather than the sign or absolute position of either variable, that has this monotonic relationship.

Because the squared coordinate differences are nonnegative, each individual axis also satisfies

$$(L_{ik} - L_{jk})^2 \leq 2(1 - \mathcal{M}_{ij}). \quad (4)$$

Consequently, two variables with normalized MI close to one cannot be widely separated on even one eigenvalue-scaled axis under the stated conditions. Their coordinates may lie on different sides of zero if both lie close to it. Conversely, separation on one axis supplies a lower bound on their full-space separation; proximity on one axis does not establish proximity after other directions are included.

PSD projection and coordinate omission have explicit effects on this identity. For any retained subset $\mathcal{K}$ of the positive eigendirections of the raw centered matrix,

$$d_{ij,\mathcal{K}}^2 = 2(1 - \mathcal{M}_{ij}) + \sum_{k:\lambda_k < 0} |\lambda_k| (v_{ik} - v_{jk})^2 - \sum_{\substack{k:\lambda_k > 0 \\ k \notin \mathcal{K}}} \lambda_k (v_{ik} - v_{jk})^2. \quad (5)$$

The first correction is introduced by discarding the negative spectrum; the second records separation lost by omitting positive coordinates. Both can vary between pairs. Accordingly, a two- or three-dimensional display is a view of the selected representation, and an apparently close pair should be checked against the fuller coordinates and the original normalized MI. The simple monotonic distance relationship is exact when both corrections vanish.

6.4.3 DISTINCTION FROM A SIGNED PCA LOADING MAP

For correlation-matrix PCA, write $\mathbf{R} = \mathbf{Q} \text{diag}(\mu_k) \mathbf{Q}^T$ and define the correlation loadings $A_{ik} = \sqrt{\mu_k} q_{ik}$. For a positive-eigenvalue component, A_{ik} is the correlation between standardized measurement i and that component's observation scores (Jolliffe and Cadima, 2016). Retaining every positive direction gives

$$\|A_i - A_j\|^2 = 2(1 - \rho_{ij}), \quad (6)$$

where $\rho_{ij} = R_{ij}$. Strong positive correlation therefore places variables close together, whereas strong negative correlation places them far apart, even when both relationships are highly informative. This is appropriate for a map of signed linear association, but measurements separated by the direction of their association can still retain equivalent information.

The Gaussian MI formula in Section 7 explains the corresponding U-shaped relationship between MI and signed correlation: under joint Gaussianity, large correlations of either sign imply high MI, and zero correlation implies zero MI. Zero correlation does not generally imply independence outside that setting. An informational-axis coordinate, by contrast, is a measurement's position in the centered variable map, not its correlation or MI with an observation-level component score.

At a fixed discrete representation, bijective relabeling of a measurement preserves its entropy and all its pairwise MI values, leaving $\mathcal{M}$ and either map geometry unchanged when the same spectral subspace is retained. A simple example is a nonconstant variable X taking values -1 and 1 and a measurement $Y = -X$: their correlation is -1 , but each completely determines the other and $\mathcal{M}_{X,Y} = 1$. Before any optional removal of redundant measurements, their full PCA correlation-loading distance is 2 . Their normalized-information profiles are identical, so the corresponding rows of $\mathcal{M}_c$ are identical. The eigenvector equation then gives $\lambda_k(v_{Xk} - v_{Yk}) = 0$: they coincide in every retained positive direction, in both Z and L . Thus neither information map separates equivalent retained information merely because its numerical coding reverses direction. For nonidentical information profiles, equal-axis and eigenvalue-scaled distances can differ because their weighting of the retained distinctions differs.

This supplies a specific rationale for using the map to explore groups defined by shared information. It does not depend on all members increasing or decreasing together. A clustering procedure based on absolute correlation could also disregard direction; the additional motivation for the information representation is sensitivity to pairwise nonlinear and categorical dependence beyond correlation magnitude.

6.4.4 FROM THE MAP TO OBSERVED-VARIABLE GROUPS

For a proposed group $\mathcal{G}$ of $g \geq 2$ measurements, let $\bar{Z}_{\mathcal{G}} = g^{-1} \sum_{i \in \mathcal{G}} Z_i$. Its equal-axis within-group dispersion is

$$W_{\text{eq}}(\mathcal{G}) = \sum_{i \in \mathcal{G}} \|Z_i - \bar{Z}_{\mathcal{G}}\|^2 = \frac{1}{g} \sum_{\substack{i < j \\ i, j \in \mathcal{G}}} d_{ij, \text{eq}}^2. \quad (7)$$

This measures agreement among the members' retained loading profiles with equal weight per axis. It can support geometric grouping without requiring a particular membership rule. For example, a specified-number k -means partition minimizes the sum of these dispersions over groups; it is one possible downstream rule, not part of the spectral diagnostic itself.

The eigenvalue-scaled criterion has a different interpretation. Let $\bar{L}_{\mathcal{G}} = g^{-1} \sum_{i \in \mathcal{G}} L_i$. In the full PSD case of Equation 3, its within-group dispersion satisfies

$$\sum_{i \in \mathcal{G}} \|L_i - \bar{L}_{\mathcal{G}}\|^2 = \frac{1}{g} \sum_{\substack{i < j \\ i, j \in \mathcal{G}}} d_{ij}^2 = \frac{2}{g} \sum_{\substack{i < j \\ i, j \in \mathcal{G}}} (1 - \mathcal{M}_{ij}). \quad (8)$$

For fixed group size, greater compactness in this full eigenvalue-scaled geometry therefore means higher mean pairwise normalized MI. With PSD projection or coordinate truncation, the corresponding terms from Equation 5 enter this dispersion as well. Equal-axis compactness instead measures agreement across equally weighted retained directions, as in Equation 7; it does not imply this exact mean-MI identity. The two criteria express different grouping objectives, so coordinate weighting should be chosen and reported before interpreting the resulting groups.

A measurement may reflect several processes even when its conventional label assigns it to one domain. For example, a symptom labeled “cardiac” may share more of its observed information profile with a cardiopulmonary group than with indicators specific to cardiac

function. Its loading profile and full-space relationships can make that discrepancy visible. The map can therefore support revision of descriptive groupings, recognition of overlapping memberships and selection of hypotheses for downstream modeling.

Entropic Scree supplies information-based representations for exploratory variable clustering: equal-axis coordinates emphasize agreement across selected informational directions, while eigenvalue-scaled coordinates emphasize their spectral weights. Automatic membership assignment additionally requires a grouping rule and its settings; neither the number of visible poles nor the selected primary rank alone determines the number of groups. A rule based only on the preserved Euclidean geometry is invariant to an orthogonal rotation of that coordinate configuration, even when the individual pole descriptions change. These are groupings of observed measurements, while identifying latent generating mechanisms remains the separate task described in Section 6.

7 Assessing Linear Representational Expansion: The Dimensional Inflation Index

A small number of generating variables can produce a much larger collection of observed measurements through nonlinear transformations, interactions, thresholds and categorical expressions. As illustrated in Section 2, these measurements can require more directions for linear representation than there are coordinates in their generating system. PCA can correctly describe this expanded linear structure, but interpreting its selected dimension as a count of generating mechanisms can consequently overstate the generating dimension. The relevant limitation is the use of linear representational dimension to answer a generative question.

Entropic Scree provides a complementary comparison by examining information relationships among measurements. Measurements that occupy different linear directions may nevertheless retain substantial shared information, allowing their overlapping dependencies to be organized within a smaller primary informational region. This motivates the directional hypothesis examined here: in nonlinear expansion systems, the dimension selected by PCA can substantially exceed the primary boundary identified by Entropic Scree. The proposed comparison is therefore intended to reveal possible linear representational expansion relative to the informational organization of the measurements.

Define the Dimensional Inflation Index as

$$\Delta_K = K_{PCA} - K_{roots},$$

where K_{PCA} is the dimension selected from the specified PCA representation and K_{roots} is the selected primary Entropic Scree boundary. Data preparation and the selection rule used for each quantity must be reported explicitly. The index quantifies the difference between these selected dimensions; interpreting that difference as expansion relative to generating dimension additionally depends on how well K_{roots} represents the generating system.

The comparison uses K_{roots} because its intended target is the primary informational organization, rather than every direction retained in the Extended Signal Tail. The latter remains relevant to the fuller description of the variable map and should be reported separately.

For jointly Gaussian scalar measurements with a nonsingular covariance matrix, population mutual information satisfies

$$I(X; Y) = -\frac{1}{2} \ln(1 - \rho^2),$$

where ρ is their Pearson correlation coefficient and information is expressed in nats (Cover and Thomas, 2006). This links Gaussian pairwise information to correlation magnitude. The nonlinear transformation, discretization, normalization and centering nevertheless produce a different spectral representation, so agreement between selected boundaries is assessed empirically.

The principal interpretations are:

- **Convergence** ($\Delta_K \approx 0$): The procedures select similar numbers of dimensions. This is compatible with a compact linear description, but does not establish that the dependence structure is linear or that the two representations capture the same relationships. Whether PCA is sufficient still depends on the intended descriptive, reconstruction or modeling objective.
- **Positive divergence** ($\Delta_K > 0$): PCA retains more directions than the primary information representation. In a system containing nonlinear expressions of shared generating inputs, this is consistent with linear representational expansion. A large difference motivates examination of whether a more compact nonlinear representation can preserve the structure relevant to the intended task. Measurement encoding, estimation effects and differences between selection rules can also contribute, so the index does not isolate an exact number of spurious dimensions or uniquely identify the cause of divergence.

Negative differences are also possible and call for examining the dependence representations, their estimation and their selection rules.

When a clear PCA elbow is present, its location can provide the specified K_{PCA} . When no clear elbow is present, that absence should be reported. An alternative criterion can still support an operational comparison, provided it is identified explicitly rather than described as an elbow. For standardized measurements analyzed through a correlation matrix, one such alternative is the Kaiser rule, $\lambda > 1$.

In the simulation reported in Section 8, the PCA Kaiser count and Entropic Scree primary boundary yield

$$\Delta_K = 5,762 - 20 = 5,742.$$

This substantial difference illustrates linear representational expansion relative to the selected informational boundary in the reported generating system.

8 Empirical Demonstration

The synthetic experiment was designed to examine a difficult distinction: can the information spectrum identify a primary boundary near the generating count when many nonlinear transformations, mixtures and noisy measurements expand the linear representation? The stated PCA, rank-PCA and RBF Kernel PCA constructions provide comparators for that

question. This is a targeted stress test, not an exhaustive performance comparison of those method families. The figures and tables describe the displayed realizations and settings; replicate selection frequencies and sampling uncertainty are not reported here.

The baseline uses $r = 20$ Gaussian generating coordinates with identity population covariance. They are independent at the generating-distribution level; finite-sample cross-correlations can deviate from zero. The known count of 20 provides a reference for the selected primary boundary while leaving open how the transformed measurements affect each representation’s spectrum.

8.1 Simulation Implementation and Reproducibility

The reported experiments use the self-contained 1.0.0 **beta** function embedded in the archived simulation script, rather than a call to the subsequently published CRAN package. The script uses the same discrete MI estimator, joint-entropy normalization, double-centering, 10^{-9} eigenvalue floor, spectral entropy and gravity formulas described above. Its automatic boundary recommendations use the following historical settings; Appendix A specifies the published version 1.0.1 settings.

- **Macro-gap reference:** Using the notation of Appendix A, the simulation begins the trailing reference at $s_{sim} = \min\{b + \max(3, \lfloor 0.05q \rfloor), q - 5\}$. It uses the largest reference gap without outlier filtering and searches all active gaps for a qualifying separation. Version 1.0.1 uses $0.10q$, filters the reference gaps at the upper three-IQR fence, and restricts macro-gap candidates to positions preceding the reference tail.
- **Extended-tail scan:** The simulation uses a fixed default window of 20 subsequent values, a fixed nominal $\alpha = 0.01$, and scans backward from J to index 1. Version 1.0.1 adapts the window to the retained measurement count, exposes **fwer_alpha**, and ends the scan at K_{log} . Both implementations set the automatic extended boundary to $\max(K_{log}, K_{tap})$ and use the same half-gap stitch and local-departure calculation.
- **Optional pole output:** The embedded function defaults to at most ten coefficients per pole, whereas version 1.0.1 defaults to the proportion described in Section 6.1. The supplied primary simulation call leaves eigenvector and bipolar-module extraction disabled, so this output-setting difference does not affect its spectra or gravity summaries.

The detector differences can change recommended boundaries even when the underlying matrix is the same. Consequently, the numerical results below are attributed to the archived simulation implementation.

The primary script sets `set.seed(19862026)`, uses the default target $\lceil N^{1/3} \rceil = 22$, and calls the embedded function with `purge_constants=FALSE`, `check_collinearity=FALSE` and `interactive_mode=FALSE`. Post-discretization duplicate removal and the default 0.05-nat entropy filter remain active. The generated measurements are complete and comprise continuous and binary columns. The same generated observed dataset is supplied to Entropic Scree and the comparators, each of which applies its specified preparation. In the archived script, `ALPHAD_PARAM=Inf` selects the identity root covariance; values 15.0, 0.75 and 0.001 select the three correlated-root settings. The ablation script uses `set.seed(42)` when

sampling continuous columns for its FD-derived target. Reproducing a reported run requires its setting, seed, retained measurement count, selected boundaries and software environment in addition to the common generator specification.

In the baseline run, all $M = 20,000$ generated measurements were retained after preparation, giving $m = 20,000$ for the reported spectral-volume calculations.

8.2 Generative Mechanics and Combinatorial Topology

The 20 coordinates supply a global pool of main effects, cross-interactions through order five, and univariate powers through degree four. The pool contains $20 + 60 + 190 + 1,140 + 4,845 + 15,504 = 21,759$ candidate terms. The reported active-term count is $K_{rlzd} = 21,759$. This describes generating functions available to the proxy construction, not 21,759 independent directions in a centered design with only 10,000 rows.

The generator creates $M = 20,000$ observable proxies before preparation; m denotes the number retained for the information-matrix analysis. Of these, 5% are reserved as pure noise, leaving 19,000 proxies constructed from the generating coordinates. Each of those proxies selects exactly 10 of the 20 roots as its blueprint (50% root inclusion). This spreads construction across many combinations without assuming in advance what network topology the resulting measurements will exhibit.

There are $B_{blue} = \binom{20}{10} = 184,756$ possible 10-root blueprints. The 19,000 proxy draws equal about 10.3% of that count, but independently sampled blueprints can repeat. The expected number of distinct blueprints is

$$B_{blue} \left[1 - (1 - 1/B_{blue})^{19,000} \right] \approx 18,056,$$

or about 9.8% of the available configurations. Thus the design spreads measurements across many possible combinations while repeatedly sampling the same 20 roots.

For two independently drawn blueprints, their number of shared roots X has the hypergeometric distribution

$$P(X = k) = \frac{\binom{10}{k} \binom{10}{10-k}}{\binom{20}{10}}, \quad k = 0, \dots, 10.$$

The expected overlap is five, and $P(3 \leq X \leq 7) \approx 0.976986$, or 97.70%. This describes substantial typical overlap. Shared roots provide opportunities for dependence, but two functions can share generating inputs without necessarily having positive pairwise MI.

Each root enters a proxy blueprint with probability $1/2$, giving expected blueprint support of $19,000/2 = 9,500$ proxies. This balances opportunities for representation across roots, although subsequent term selection, coefficients and noise determine the observable contribution.

A 10-root blueprint has a local menu of 667 terms: 10 main effects, 30 additional powers, and 45, 120, 210, and 252 interactions of orders two through five, respectively. The implementation samples 67 terms, the rounded value of 0.10×667 , and assigns normally distributed weights with mean zero and standard deviation one. High-order terms occupy most of the menu, but a particular high-order term is eligible in relatively few blueprints.

Every expansion column is centered and standardized to unit sample variance before proxy construction. The active-term count includes terms with at least one nonzero proxy

coefficient. K_{eff} is computed from the covariance spectrum of the active standardized expansion columns, using the configured term-scale weights, all of which equal one here. The code retains covariance eigenvalues above 10^{-9} for that entropy calculation. Thus K_{eff} describes the active generating-term design before its random coefficients combine terms into observed proxies.

Table 2: **Candidate Term Pool and Local Menu.** Counts and proportions for the global 20-root pool and a selected 10-root blueprint. Each local blueprint contains 667 candidate terms.

Term Type	Total in Global Candidate Pool (21, 759)	Share of the Local Candidate Menu
Main Effects (1D)	20 (0.09%)	1.50% (10 terms)
Additional Univariate Powers (Degrees 2–4)	60 (0.28%)	4.50% (30 terms)
2-Way Interactions	190 (0.87%)	6.75% (45 terms)
3-Way Interactions	1, 140 (5.24%)	17.99% (120 terms)
4-Way Interactions	4, 845 (22.27%)	31.48%(210terms)
5-Way Interactions	15, 504 (71.25%)	37.78%(252terms)

A specified main effect has expected representation $19,000(1/2)(67/667) \approx 954$ proxies. A specified five-root interaction first requires all five roots to appear, with probability $\binom{15}{5}/\binom{20}{10} \approx 0.016254$, giving expected representation of about 31 proxies. This difference helps motivate the hypothesis that lower-order contributions may be more repeatedly expressed. It does not by itself determine how many spectral directions those contributions generate.

The experiment generates $N = 10,000$ observations for $M = 20,000$ measurements ($M > N$). Each nonconstant deterministic proxy score is standardized to unit sample variance after the selected expansion terms have been combined; pure-noise proxies have zero deterministic score. Independent Gaussian perturbations with variance $1/10$ are then added at the proxy-score level. The script calls this first stage *structural uncertainty* and sets `structural_snr=10`. The generating root values themselves remain unchanged.

Let Φ denote the standard-normal cumulative distribution function and Φ^{-1} its quantile function. For each binary proxy, the standardized score first receives an offset $\sqrt{2}\sigma\Phi^{-1}(u_j)$, where u_j is sampled uniformly from $[0.1, 0.9]$ and $\sigma = \sqrt{1 + 1/10}$. After the Gaussian perturbation, the resulting score s_{aj} is mapped to a probability $p_{aj} = \Phi(s_{aj}/\sigma)$, and a binary value is sampled from $\text{Bernoulli}(p_{aj})$. The offset varies the binary base-rate setting across measurements; the achieved prevalence also depends on the generated score distribution.

For the second measurement-error stage, a fresh vector of independent standard-normal errors is drawn for each continuous manifestation. Each error vector is then rescaled so that its sample variance equals half that manifestation’s sample variance (continuous SNR = 2), and added to the manifestation. Each sampled binary value is independently flipped with probability 0.15. The binary construction therefore includes Bernoulli sampling variation as well as the specified bit-flip error.

8.3 Behavior of the Evaluated Baselines

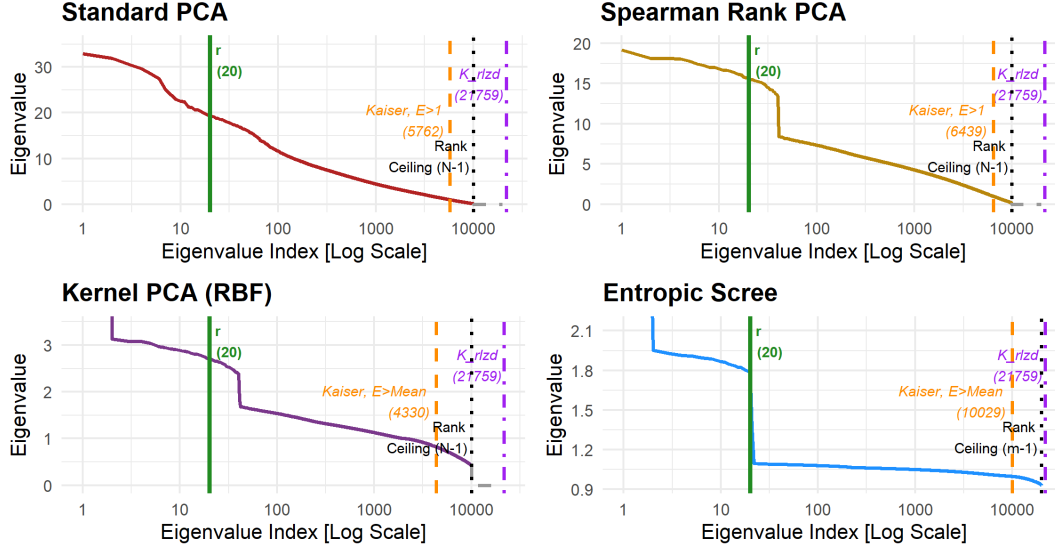


Figure 1: **Baselines and Entropic Scree.** The comparison uses a logarithmic eigenvalue-index axis and a linear eigenvalue axis. The reference line at 20 marks the known generating count. Standard PCA has no clearly resolved elbow at that count in the reported spectrum. Rank-PCA and the specified RBF Kernel PCA show a feature near index 40 with an extended tail. Entropic Scree selects a primary boundary at 20. Possible relationships between the index-40 feature, polynomial transformations and interaction structure are discussed as interpretations.

Standard PCA: The script applies `prcomp` with centering and unit-variance scaling, corresponding to a sample correlation representation. Its centered data matrix has rank at most $N - 1 = 9,999$. That limits the linear directions supported by this sample, even though the generating construction contains 21,759 active terms. As described in Section 2, nonlinear functions of the same root can require different linear directions. The reported PCA spectrum has no comparably resolved elbow at the generating count of 20, and the Kaiser rule retains 5,762 components. It therefore gives a much larger selected linear representation in this constructed system; these retained directions should not be read as independent generating causes.

Spearman Rank PCA: The script replaces each column by its ranks using `data.table::frank` with default average ranks for ties, then applies centered, unit-variance-scaled PCA. Ranking removes strictly increasing marginal transformations under consistent tie handling. For a symmetric continuous X , pure odd powers preserve its ordering, whereas X^2 folds positive and negative values together. Consequently, X and X^2 can have zero rank correlation, even though X^2 and X^4 have identical ordering. This supplies a useful illustration of why rank transformation can remove some nonlinear expansion while retaining distinctions between monotone and folded responses.

The simulated proxies, however, are noisy mixtures of many terms. Ranking a sum is not the same as ranking and then summing each term. The pure-transform example therefore motivates, but does not prove, the interpretation that the feature near 40 reflects distinct monotone and folded patterns associated with 20 roots. Establishing that explanation would require comparing the extracted directions with the known generating functions.

In the reported spectrum, the eigenvalue after that feature is approximately 8.0, followed by an extended decline. A visual selection near 40 is twice the generating count, while the Kaiser rule retains 6,439 directions. The continuing tail makes the retained dimension sensitive to the selection rule.

Kernel PCA (RBF): The kernel provides a nonlinear similarity between records. In this implementation, measurements are standardized and $K_{ab} = \exp[-\gamma\|x_a - x_b\|^2]$ with $\gamma = 1/m$; the resulting $N \times N$ kernel is centered to give K_c . The reported spectrum has a feature near 40, a following eigenvalue above 1.6, and 4,330 directions above its mean eigenvalue. This shows that the specified nonlinear representation and rule also do not select 20 in this experiment.

A correspondence with the odd/even distinction is one possible explanation of the visual similarity to rank-PCA, but it does not follow automatically from the RBF formula. Kernel bandwidth, scaling, noise and the mixture structure also influence the spectrum. The different absolute eigenvalue scales across these representations are not directly comparable signal calibrations.

The descriptions of visual elbows and the Kaiser-based Δ_K are the manuscript’s stated comparisons. The simulation script additionally prints a maximum successive log-gap for each baseline, excluding the first gap. That numerical helper returns a candidate when the search has enough positive eigenvalues; it does not determine whether a distinct visual elbow exists. Its candidates should therefore be distinguished from the visual judgments and Kaiser counts reported here.

8.4 Entropic Scree Results and Combinatorial Interpretation

For the same simulated dataset, Entropic Scree selects a primary boundary matching the known generating count of 20. The extended region and gravity summaries provide additional descriptions of the resulting map.

Centering changes the unit-diagonal trace from $m = 20,000$ to the reported $\text{Tr}(\mathcal{M}_c) = 19,945.14$. The Negative Spectral Mass Ratio (NSMR), expressed as a percentage, rounds to 0.000%, indicating very little discarded negative spectral mass in this run.

The Log-Gap recommendation selected $K_{roots} = 20$. The next eigenvalue was approximately 1.1. The Triple-Tap recommendation placed $K_{extended} = 211$, with a reported departure of 39.11 local regression-residual standard deviations. That is the detector’s descriptive departure statistic, not a calibrated Gaussian significance level.

The location of the extended boundary motivates a combinatorial interpretation. The generator has 20 main effects and $\binom{20}{2} = 190$ pairwise interaction terms, totaling 210. Particular low-order terms have more opportunities to recur across proxies than particular high-order terms, as shown in Section 8.2. Repeatedly expressed distinctions may therefore contribute more visibly to the extended spectrum. This offers a plausible explanation to investigate for a boundary near 210, while preserving the reported index 211.

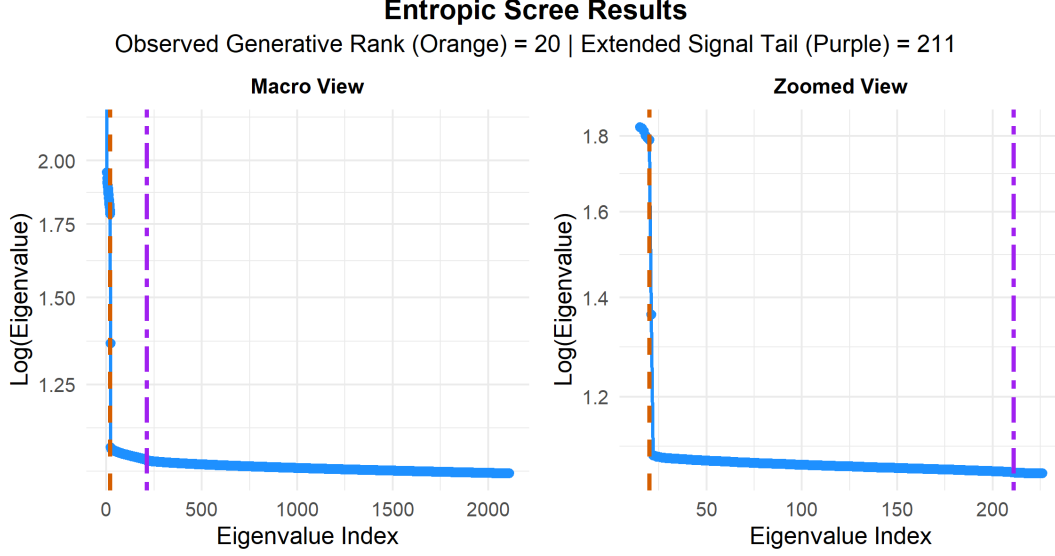


Figure 2: **Entropic Scree Extraction.** Both panels display eigenvalues on a logarithmic vertical axis against a linear eigenvalue index. The macro view shows the broader spectral context, and the zoomed view emphasizes the recommended primary and extended boundaries. In the reported non-interactive run, these give $K_{roots} = 20$ and $K_{extended} = 211$. R_{eff} is a scalar summary of the normalized working spectrum.

The numerical proximity alone does not identify 190 eigenvectors with the 190 interaction terms or show that higher-order terms lie entirely beyond the boundary. Here $K_{extended} = 211$ includes the 211th axis in the signal allocation; the boundary is therefore near, rather than exactly equal to, the count of 210 main and pairwise terms. A targeted test would remove selected interaction orders, or compare the known term responses with extracted directions.

The reported $R_{eff} = 19,963.87$ and selected extended boundary allocate 1.451% of the measurement-count reference to total shared signal and 98.549% to idiosyncratic informational volume. These are the proposed spectral allocations, not a direct measurement that 98.549% of the observed data values consist of random error. The generator’s SNR settings describe a different quantity.

AIG is 14.512 variable equivalents per primary axis, while $FSIG_1 = 74.546$ variable equivalents is the allocation to the leading axis. The profile $[1.000, 0.160, 0.158, 0.157, 0.156, \dots, 0.146]$ shows one strongly weighted informational axis followed by a more evenly weighted remainder. In the context of the balanced blueprint design, a broad pattern of shared structure together with more balanced secondary distinctions is a reasonable hypothesis. A complementary explanation to examine is separation associated with measurement type, discretization or finite-sample estimation. Inspecting the leading loadings by measurement type and comparing with independently column-permuted data would help distinguish these

possibilities. Such a comparison is proposed here, rather than reported as an established result.

The central finding remains that the information diagnostic selected 20 in a setting where the evaluated baseline selections did not. The PCA Kaiser comparison yields $\Delta_K = 5,742$; visual selections near 40 from the other representations are twice the generating count. These are substantial differences in the stated experiment. The profile and tail patterns additionally motivate testable explanations of the observed spectral organization.

Table 3: **Empirical Simulation Summary.** Comparison of dimensional diagnostics extracted from a noisy, high-dimensional ($m > N$) mixed-data environment.

Metric	Observed Value
<i>Panel A: Environment Configuration</i>	
Sample Size (N)	10,000
Intrinsic Generative Rank (r)	20
Generated Observable Proxies (M)	20,000 (80% Continuous, 20% Binary)
Retained Measurements (m)	20,000
Independent Measurement Error Injected	SNR = 2.0 (Cont.), 15% Bit-flip (Binary)
Proxy-Score Structural Perturbation	Gaussian variance 0.1 (nominal SNR = 10)
Centered Sample Covariance Rank Ceiling	9,999
Realized Intrinsic Root Expansion Rank (K_{rlzd})	21,759
Effective Intrinsic Root Expansion Rank (K_{eff})	3,519.33
<i>Panel B: Standard PCA Diagnostics</i>	
Kaiser Criterion: $\lambda > 1.0$	5,762
Visual Elbow	None identified; Kaiser count used for Δ_K
<i>Panel C: Spearman Rank PCA Diagnostics</i>	
Kaiser Criterion: $\lambda > 1.0$	6,439
Visual Elbow	~ 40 (Ambiguous; obscured by elevated tail)
<i>Panel D: Kernel PCA (RBF) Diagnostics</i>	
Mean-Kernel-Eigenvalue Rule: $\lambda > \text{Tr}(K_c)/N$	4,330
Visual Elbow	~ 40 (Ambiguous; obscured by elevated tail)
<i>Panel E: Entropic Scree Diagnostics</i>	
Centered Trace ($\text{Tr}(\mathcal{M}_c)$)	19,945.14
Negative Spectral Mass Ratio (NSMR; reported as %)	0.000%
Mean-Working-Eigenvalue Rule: $\eta_i > S_\varepsilon/m$	10,029
Observed Generative Rank (K_{roots})	20
Extended Signal Tail boundary ($K_{extended}$)	211
Total Unique Probabilistic Volume (R_{eff})	19,963.87
Idiosyncratic Informational Volume (% of m)	98.549%
(Residual spectral allocation; not measured error fraction)	

Continued on next page

Table 3 continued

Metric	Observed Value
Total Shared Signal Volume (TSV ; % of m)	1.451%
Automated Extraction Method	Dual-Diagnostic Ensemble
Primary-Boundary Eigenvalue Ratio	$1.31 \times$ (23.8% Drop)
Triple-Tap Departure / Local Residual SD	39.11
Average Informational Gravity (AIG)	14.512 Variable Equivalents per Primary Axis
Leading-axis FSIG ($FSIG_1$)	74.546 Variable Equivalents
Remaining primary-axis FSIG ($FSIG_i$, $i = 2, \dots, 20$)	$\sim 11.91 - 10.92$ Variable Equivalents
Structural Topology Profile (T_i , $i = 1, \dots, 20$)	[1.000, 0.160, $\dots$, 0.146] (Dominant First Axis; Flatter Remaining Profile)
Dimensional Inflation Index (Δ_K)	5,742 (PCA Kaiser Count Minus K_{roots})

8.5 Robustness to Root Entanglement

The independent-root baseline isolates a particular generating configuration. Real applications can involve dependent processes, so the same framework was also examined under the specified correlated-Gaussian-root settings. These runs test a change in the generator, not every possible form of root dependence.

As described in Appendix C, the rank-PCA and RBF-kernel features near 40 were no longer comparably resolved in the correlated-root runs. Entropic Scree retained a transition near the generating count, with $K_{roots} = 21$ in the highest setting. This is a one-axis departure from the nominal count of 20, supporting approximate stability in that setting rather than exact recovery.

8.6 Illustrative Mapping of Symptom Measurements

To illustrate the measurement geometry, Entropic Scree was applied to Chhetri’s publicly available *Mental Health Symptoms & Illness Prediction* dataset (Chhetri, n.d.). The source describes these data as synthetically generated or enhanced from DSM-5-inspired criteria for educational use. The analyzed file contained $N = 8,304$ records, 185 binary symptom measurements and a categorical **Disease** label. The label was excluded, leaving $m = 185$ measurements. The binary states were treated as categorical, and all 185 symptom measurements were retained. The analysis used the published **Entropic.Scree** version 1.0.1, with the optional collinearity filter disabled.

The selected primary rank was $K_{roots} = 20$, retaining 20 Informational Axes for the primary measurement map. Figures 3 and 4 display the same measurements using equal-axis coordinates (v_{i1}, v_{i2}) and (v_{i1}, v_{i2}, v_{i3}) , respectively. These views show only the first two and first three of the 20 retained primary axes, without eigenvalue scaling; neither displays the full 20-dimensional primary configuration. Each point represents a symptom measurement, not a record or an extracted latent root.

The two-axis map separates concentrations containing *vomiting*, *fidgeting* and *blackouts*, while many other measurements remain close in that projection. Adding Axis 3 resolves further separation within the central concentration. For example, *delusion* and *fatigue* are

separated by 0.0137 in the two-axis map but by 0.0971 in the three-axis map, because their third-axis loadings differ substantially. This illustrates how joint loading coordinates reveal distinctions that a single pole or a lower-dimensional view can conceal. Proximity here concerns the displayed equal-axis coordinates; omitted axes can reveal additional separation. The example demonstrates exploratory measurement mapping in a synthetic symptom dataset, rather than validating clinical groupings or estimating latent causes.

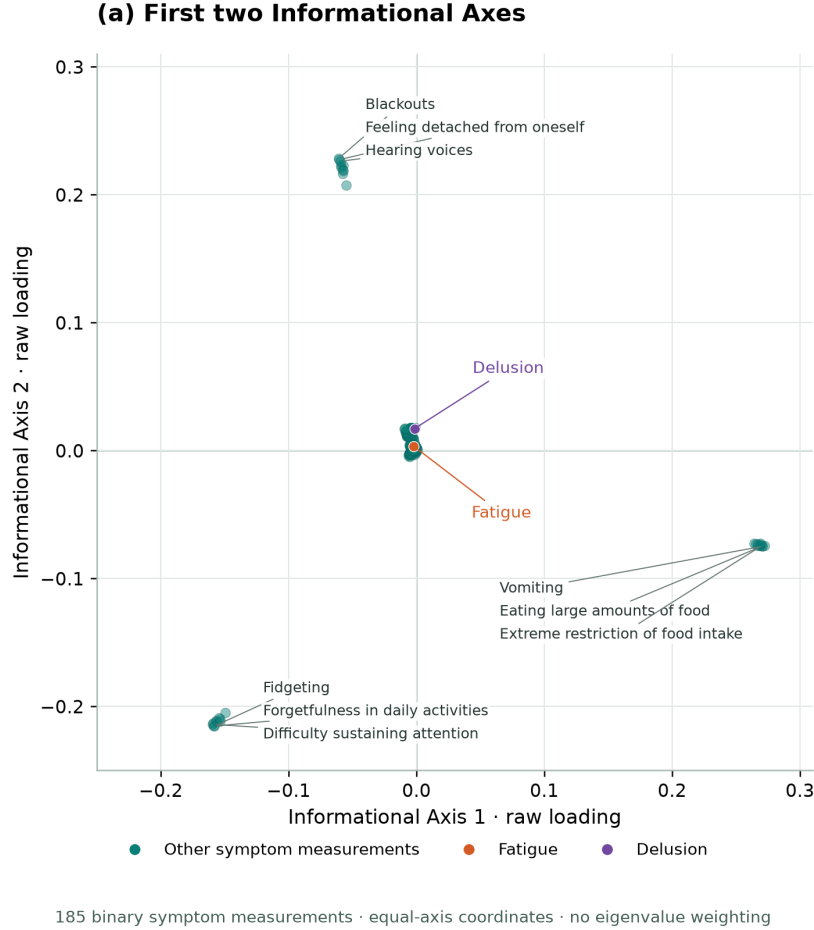
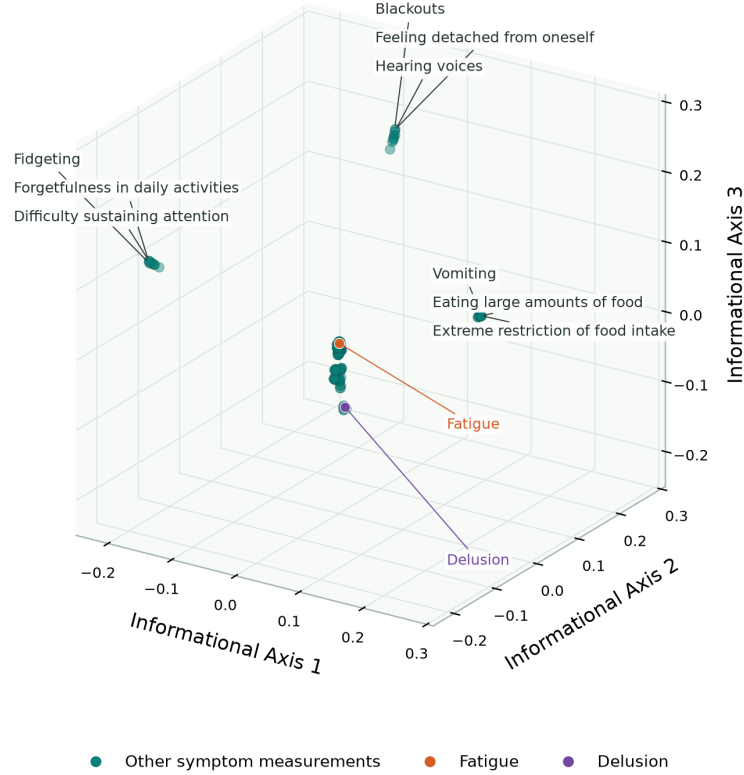


Figure 3: **Symptom measurements on the first two Informational Axes.** All 185 retained binary symptom measurements are plotted at their raw, signed eigenvector coordinates. Selected measurements are labeled for orientation. Orange and purple identify *fatigue* and *delusion*, respectively; teal denotes the remaining measurements, not an assigned cluster. Marker sizes are uniform apart from slight highlighting of the two examples. Several measurements overlap or lie very close together. The excluded **Disease** label did not contribute to the map.

(b) First three Informational Axes

185 binary symptom measurements · equal-axis coordinates · no eigenvalue weighting

Figure 4: **The same symptom measurements on the first three Informational Axes.** Axis 3 reveals additional separation among measurements that appear close in the two-axis view, including the highlighted *fatigue* and *delusion* pair. Colors have the same meaning as in Figure 3. An orthographic viewing angle is used to display the three coordinates; no additional projection of all retained axes is performed. Distances refer to this three-axis equal-weighted configuration, not to the complete retained space or directly to normalized variation of information.

9 Conclusion

Entropic Scree extends the diagnostic use of information-matrix eigenanalysis by organizing its spectrum and measurement coordinates into a common interpretive framework. Earlier MI-based feature extraction, information-distance mapping and clustering establish the broader approach (Benjaminsson et al., 2010; Tong and Shi, 2015; Jiang and Yan, 2015; Lv et al., 2021; Pan et al., 2025). Here, joint-entropy normalization places the represented pairwise dependencies on a relative-information scale, and centering with positive-spectrum

projection produces a measurement map with explicit geometric interpretation. The primary and extended boundaries address how much of that representation to retain and how its selected spectral regions enter the gravity accounting.

The framework connects three levels of description: the selected primary and extended spectral regions, the organization of measurements across individual poles and their joint coordinate space, and the allocation of spectral volume through AIG and FSIG. The gravity metrics distinguish the overall shared-signal percentage from the shared informational footprint allocated to the primary axes. Consequently, dominance of idiosyncratic informational volume does not, by itself, imply a negligible shared footprint for the selected primary structure. Equal-axis mapping supports exploratory variable clustering by giving retained informational directions equal overall geometric weight, making secondary distinctions easier to examine alongside dominant contrasts. Its joint distances depend on the retained subspace and are unchanged by orthogonal changes of basis within that subspace. Eigenvalue-scaled mapping provides a complementary description: with a PSD centered matrix and all positive coordinates retained, squared distances equal twice normalized variation of information, with projection and truncation effects given in Section 6.4. Both representations preserve invariance to bijective recoding of represented states, allowing information-sharing groups to be examined without requiring a common direction of numerical association.

In the nonlinear, noisy mixed-data experiment, the selected primary boundary matched the 20 generating coordinates, whereas the evaluated PCA, rank-PCA and RBF-kernel selections differed. The primary selection remained 20 across the three tested bin targets even though gravity estimates changed markedly. Correlated-root experiments retained a boundary near the generating count. These results support the diagnostic in the configurations studied and motivate broader investigation.

The distinction between sample depth and measurement breadth remains important. Additional observations can improve estimation of existing measured relationships; they do not automatically remove measurement error inherent in each observation. Additional informative measurements can make previously indistinguishable process configurations more observable, consistent with the measurement-breadth motivation in G2G (Lee-St. John et al., 2026). The benefit depends on what the added measurements reveal, not on column count alone. The intended output is an operational map of resolved informational distinctions, which may differ from the nominal count of generating parts.

The construction is nonparametric in the sense that it does not require a specified parametric family for the generating distribution. Its interpretation depends on the pairwise representation, discretization, estimation, PSD projection and boundary selection.

Future work should vary sample-to-feature ratios, dependence structures, measurement breadth, noise levels, category balance and discretization; include replicated simulations and explicitly tuned comparators; and assess stability in substantive datasets. Comparisons within the MI-matrix family should distinguish shared construction choices from task-specific objectives: MI-based projections are relevant to downstream feature representations, whereas MI-spectral clustering provides a comparator for observed-variable grouping (Tong and Shi, 2015; Jiang and Yan, 2015; Lv et al., 2021). Holding the estimated pairwise information fixed while varying normalization, centering, PSD treatment and spectral-boundary rules would help isolate which choices account for changes in the diagnostic.

The multiaxis interpretation also motivates direct evaluation of variable groupings under reversed coding, nonlinear dependence and mixed measurement types. Equal-axis and eigenvalue-scaled coordinates should first be compared at the same retained axis set and with the same membership rule, then examined across alternative retained ranges. Known or independently supported groups can be used to assess membership recovery and stability, while distortion of the original information dissimilarities measures a separate objective. Comparators should include direct NVI-based grouping, MDS-based information-distance maps (Benjaminsson et al., 2010; Pan et al., 2025), MI-spectral clustering, signed PCA loading maps and absolute-correlation alternatives. For classical MDS, the choice between D_{ij} and $\sqrt{2D_{ij}}$ as input distances should be specified, because the latter reproduces the centered operator used here before the stated spectral treatment and coordinate weighting. These comparisons would establish when equal weighting reveals meaningful secondary structure and when it emphasizes unstable directions. Genomic and clinical settings are potential applications where mixed measurement forms and nonlinear organization make these questions relevant. The current experiments establish the reported outcomes against PCA, rank-PCA and the specified RBF kernel; comparative clustering performance within the MI-method family remains to be evaluated. Stability of the selected rank and sensitivity of the gravity estimates should also be investigated separately.

Code Availability

The R implementation is published on CRAN as `Entropic.Scree`. Version 1.0.1 was published on 15 September 2026 and is distributed under the Apache License 2.0. The package, reference manual and installation information are available at <https://CRAN.R-project.org/package=Entropic.Scree>; its persistent identifier is <https://doi.org/10.32614/CRAN.package.Entropic.Scree>. It can be installed in R with `install.packages("Entropic.Scree")`. The package provides `Entropic.Scree()` and the companion `Update.Entropic.Scree()`.

The simulation code is available separately on GitHub. The archived main script for Section 8 and Appendix C is available at <https://github.com/tjleestjohn/Entropic-Scree/blob/064dcbd1401633c16669fe3d9826c6b37a9fe2d8/Entropic.Scree.R.Simulation%20-%20ENLI.R>.

The companion binning-ablation script for Appendix B is available at <https://github.com/tjleestjohn/Entropic-Scree/blob/064dcbd1401633c16669fe3d9826c6b37a9fe2d8/Appendix.B.Binning.Ablation.R>.

These links identify the reviewed simulation revision. The main script contains its historical function definition, and the ablation script reuses that definition, dataset and baseline result. Section 8.1 records its differences from the published package.

Computational Efficiency and Scaling

The implementation reduces repeated work in constructing a dense pairwise information matrix. Three choices contribute to its practical computational behavior:

1. **Preprocessing:** Constant, duplicate and collinearity checks can reduce redundant work, and the supplied R code also checks for identical discretized columns. A duplicate

after binning is not necessarily a duplicate at the original resolution. The supplied primary simulation script disables the initial constant/duplicate and collinearity options.

2. **Discrete frequency calculation:** Continuous measurements are represented by equal-frequency categories, allowing marginal and joint entropies to be calculated from counts. This avoids fitting a continuous density for every pair. The default target scales approximately as $N^{1/3}$, balancing a manageable number of cells against retained resolution. The bin-count target is an empirical tuning choice examined in Appendix B.

Coarser bins can discard within-bin distinctions, whereas finer tables can become sparsely sampled. Normalization puts pairs on a relative-information scale but does not guarantee proportional cancellation of numerator and denominator biases. Appendix B therefore evaluates discretization empirically, separating selected-rank behavior from changes in volume and gravity.

3. **Compiled computation and parallelism:** The published package implements pairwise count calculations in a compiled Rcpp/C++ backend, using OpenMP where supported by the build and platform. The historical simulation script instead compiles its embedded backend at runtime and provides a serial fallback when its OpenMP compilation attempt fails. Both arrangements reduce interpreter overhead; parallel builds additionally distribute pair calculations across processor threads.

The implementation retains dense-matrix scaling. Its pairwise count backend performs $O(m^2(N + B_{bins}^2))$ work, followed by a dense eigendecomposition that generally requires $O(m^3)$ time. Dense matrices require $O(m^2)$ storage, alongside the prepared $N \times m$ data and temporary count tables. A single 20,000 $\times$ 20,000 double-precision matrix occupies approximately 3.2 GB in decimal units; total memory demand is larger because several arrays and eigensolver workspaces can coexist. Parallel execution reduces elapsed time but does not change these storage requirements.

Data Availability Statement

The simulation experiments use synthetic data generated by the simulation scripts available at <https://github.com/tjleestjohn/Entropic-Scree>. These scripts specify the simulation settings and random seeds and provide the procedures for generating the datasets and calculating the numerical results.

The separate symptom-mapping illustration in Section 8.6 uses Chhetri’s publicly available synthetic educational dataset, *Mental Health Symptoms & Illness Prediction*, available at <https://www.kaggle.com/datasets/sarveshchhetri/mental-health-symptom-dataset-one-hot-encoded>.

Details of the software implementations are provided in the Code Availability section.

Declaration of Competing Interests

The author is affiliated with Enli. The methodology and codebase introduced in this manuscript are provided entirely open-source for the scientific community. This research

received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors, and the author declares no other competing financial or personal interests.

AI Use Disclosure

AI tools assisted with language copyediting and identifying potentially relevant literature. The author reviewed the cited sources and takes responsibility for the manuscript’s content.

Appendix A. Automated Extraction Heuristic Mechanics

This appendix describes the Dual-Diagnostic Ensemble in the published `Entropic.Scree` version 1.0.1. It supplies a primary and an extended boundary from the same spectrum. A preliminary macroscopic search region limits the range in which smaller local irregularities compete with those transitions. Section 8.1 records the settings of the historical simulation implementation.

Phase 1: Macroscopic search region. Let η_i be the ordered working eigenvalues from Section 3.4, and let $\mu = \text{Tr}(\mathcal{M}_c)/m$ use the unfloored centered trace. Define $q = \#\{i : \eta_i > 10^{-8}\}$ active values and $b = \#\{i : \eta_i > \mu\}$. When $q > 10$, form $g_i = \eta_i - \eta_{i+1}$ for $1 \leq i < q$ and start the trailing reference at

$$s = \min\{b + \max(3, \lfloor 0.10q \rfloor), q - 5\}.$$

Let Q_1 and Q_3 be the first and third quartiles of $\{g_i : s \leq i < q\}$, calculated using R’s default sample-quantile convention. Retain the reference gaps satisfying

$$g_i \leq Q_3 + 3(Q_3 - Q_1).$$

Their maximum defines g_{ref} ; the code uses the reference median if the filtered set is empty. Filtering prevents an unusually large secondary cliff in the trailing region from setting the entire reference scale. The qualifying threshold is

$$\delta = \max(10^{-6}, 1.5g_{ref}).$$

Only indices before that trailing reference are considered as macro-gap candidates. If any qualify, define

$$j_{bulk} = 1 + \max\{i : 1 \leq i < s, g_i > \delta\}, \quad J = \max(2, j_{bulk} - 1).$$

When no macro-gap is identified, including the short-spectrum path, the search endpoint defaults to $J = b$.

Phase 2A: Primary recommendation by a secondary log-gap. On the standard path, select the largest eligible gap

$$K_{log} = \underset{2 \leq i \leq \min(J, q-1)}{\operatorname{argmax}} [\ln(\eta_i) - \ln(\eta_{i+1})].$$

The logarithm makes this a ratio comparison. Excluding $i = 1$ prevents a dominant leading axis from automatically determining the recommendation. If there is no standard candidate,

the corresponding fallback returns one. For $q < 3$, the code returns $\max(1, b)$. When a detected bulk begins before index 4, a separate low-rank path examines the second-largest active log-gap and uses $\max(1, j_{bulk} - 1)$ if that gap is already within the bulk. Thus low-rank paths can return one.

Phase 2B: Extended recommendation by the Triple-Tap local trajectory-departure heuristic. If a macro-gap was found, add half its magnitude to every working eigenvalue at or beyond j_{bulk} . This “stitch” changes the trajectory examined by the detector so that the macro discontinuity does not alone dominate the local regression. The original working eigenvalues remain the inputs to the volume calculations.

For integer-converted requested window w_0 (default 20), the published window is

$$w = \max\{5, \min(w_0, \lfloor 0.10m \rfloor)\}.$$

When $J \geq 3$, scan backward from J through K_{log} . At each candidate i , fit a line to the next w stitched eigenvalues, using reference positions $0, \dots, w - 1$. Extrapolation to position -1 gives the expected candidate value $\hat{y}_i$; the residual scale is $s_i = \sqrt{\text{RSS}/(w - 2)}$. Both are bounded below by 10^{-12} for numerical safety. With $\alpha = \text{fwer_alpha}$ (default 0.01), set

$$c_i = \max\{2, t_{1-\alpha/J, w-2}\}.$$

The first candidate encountered with stitched value greater than $\hat{y}_i + c_i s_i$ gives $K_{tap} = i$. If only ℓ subsequent values remain, with $2 \leq \ell < w$, use their mean and sample standard deviation, with $\ell - 1$ degrees of freedom. If no departure is found, or the scan is unavailable, set $K_{tap} = K_{log}$.

The automatic recommendations are

$$K_{roots} = K_{log}, \quad K_{extended} = \max(K_{log}, K_{tap}).$$

Both are inclusive retained-axis indices. The published interface allows interactive revision of these recommendations.

The adjusted quantile sets a nominal threshold for this adaptive search. Its family-wise false-positive rate has not been established as α . Calibration would need to account for the adaptive selection and the use of regression residual standard deviation rather than the full prediction standard error.

Appendix B. Robustness to Discretization (Binning Ablation Study)

Discretization determines which distinctions are visible to the empirical information estimator. Histogram sensitivity has also motivated alternative MI-matrix estimators: Contreras-Reyes (2021) studied parametric skew-elliptical models as an alternative to histogram-based time-series information analysis. Here, coarse categories combine nearby values and can discard dependence expressed within bins. Finer categories preserve more distinctions but leave fewer observations per joint cell. Plug-in entropy is generally downward biased, whereas plug-in MI commonly has positive bias near independence (Paninski, 2003). Normalized information depends on both the numerator and denominator, so its response to changing bin count can be nonmonotonic. The following ablation examines how this estimator choice affects boundary selection and gravity separately within the present framework.

For a continuous – continuous pair with B_{occ} occupied marginal bins, there are up to B_{occ}^2 joint cells. At $N = 10,000$, the 22-, 100- and 208-bin targets correspond to averages of approximately 20.66, 1.00 and 0.231 records per possible joint cell, respectively. Actual occupancy is uneven and depends on the relationship. The finer targets therefore alter both the represented distinctions and the sampling regime; increased allocated gravity cannot be attributed to recovered dependence alone. Independently permuting each column while preserving its marginal counts would provide a useful finite-sample reference.

The ablation therefore asks whether changing resolution changes the selected primary boundary, the continuous spectral summaries, or both. It uses the same synthetic system with three bin targets:

1. $N^{1/3}$ **target (22 bins)**: The lower-resolution equal-frequency setting used in the primary experiment.
2. $N^{1/2}$ **target (100 bins)**: A finer equal-frequency setting used to test sensitivity to increased cardinality and sparsity.
3. **Freedman–Diaconis-derived target (208 bins)**: In the ablation script, `nclass.FD` is evaluated on up to 100 sampled columns with more than 10 unique values. Their proposed counts are averaged and rounded to obtain a global target for the same equal-frequency discretizer. This is a change of bin target, not ordinary variable-specific equal-width Freedman–Diaconis binning.

The comparison uses the same complete $10,000 \times 20,000$ observed dataset and the historical function described in Section 8.1. The companion script reuses the original 22-bin result, then evaluates the 100-bin and FD-derived targets. Each call disables the initial purge, collinearity check and interactive override, while retaining the post-discretization and entropy filters. Because the filters are reapplied at each resolution, both the retained count m and the selected $K_{extended}$ enter the comparison. The gravity differences reflect the full preparation and boundary-selection pipeline, rather than a change in bin target with every other numerical quantity held fixed.

Table 4: **Discretization Ablation Results.** The selected primary rank was 20 across the three tested targets. Effective spectral volume, allocated shared signal and gravity estimates changed substantially, demonstrating that stable rank selection can coexist with substantial changes in volume and gravity estimates.

Discretization Rule	$N^{1/3}$	$N^{1/2}$	FD-derived target
Global Bin Target	22	100	208
Total Unique Probabilistic Volume (R_{eff})	19,963.87	18,996.07	16,727.77
Total Shared Signal Volume (% of m)	1.45%	6.21%	38.07%
Observed Generative Rank (K_{roots})	20	20	20
Average Informational Gravity (AIG)	14.51	62.12	380.69
Leading Axis FSIG ($FSIG_1$)	74.55	1,068.64	7,298.46

Table 4 shows two different outcomes. The selected primary count stayed at $K_{roots} = 20$, while R_{eff} decreased and the shared allocation and gravity estimates increased substantially. A boundary identified from relative separation can remain in the same place even when much of the spectrum changes. This is the useful empirical stability demonstrated here.

Relative log-gaps are unchanged by a common positive rescaling of the eigenvalues:

$$\ln(c\lambda_i^+) - \ln(c\lambda_{i+1}^+) = \ln(\lambda_i^+) - \ln(\lambda_{i+1}^+), \quad c > 0,$$

for positive adjacent eigenvalues. This explains why relative separation can be examined independently of absolute spectral scale. Changing binning can also alter relative eigenvalue weights, so the ablation assesses whether the primary separation persists under those changes. This invariance applies to the log-gap calculation itself; the complete implementation also contains fixed numerical floors and absolute thresholds that can matter when a rescaling crosses them.

Eigspectra Across Discretization Regimes

Generative signal ($K=20$) remains invariant
despite artificial compression of the topological space

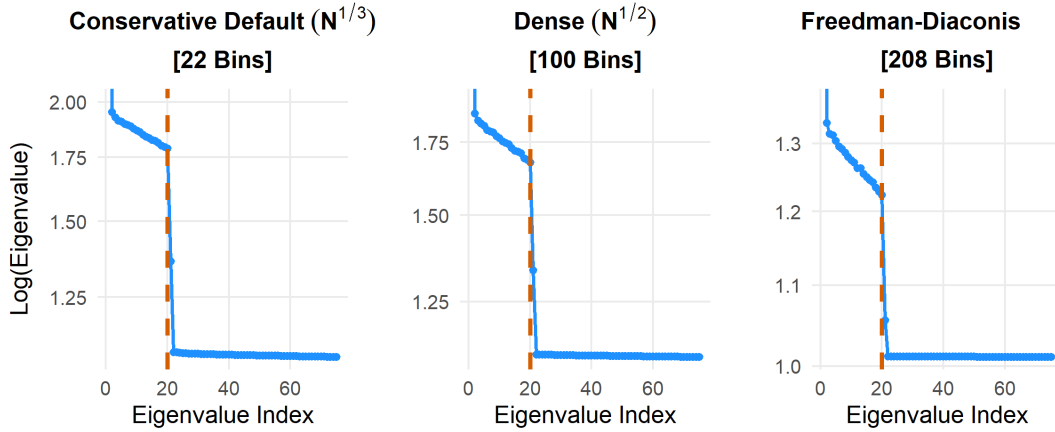


Figure 5: **Eigspectra across Discretization Targets.** The 22-, 100- and 208-bin settings yield different spectral distributions while retaining a reported primary selection of 20. Eigenvalues are displayed on a logarithmic vertical axis. The dashed line marks the known generating count, $r = 20$; the selected ranks are reported separately in Table 4. The plots show the observed changes; they do not separately identify estimation bias, sampling variation and changes in the discretized population representation. The upper display limit is set to $1.025\eta_2$ to emphasize the secondary spectrum, so the leading eigenvalue can lie above the visible range.

The sensitivity of gravity can be understood directly from the accounting. For fixed m ,

$$TSV = m - R_{eff}(1 - P_{sig}), \quad \left. \frac{\partial TSV}{\partial R_{eff}} \right|_{P_{sig}} = -(1 - P_{sig}).$$

Thus a larger effective rank lowers TSV if the selected spectral fraction is held fixed. But changing bins can also change that fraction. The total differential is

$$dT_{SV} = -(1 - P_{sig}) dR_{eff} + R_{eff} dP_{sig}.$$

Changes in both R_{eff} and P_{sig} therefore contribute to the change in TSV. FSIG can additionally change because each axis's proportional primary weight changes.

Interpreting differences in volume and gravity requires accounting for measurement preparation, resolution, estimation and boundary choices. These settings should accompany reported gravity values.

Appendix C. Robustness to Root Entanglement

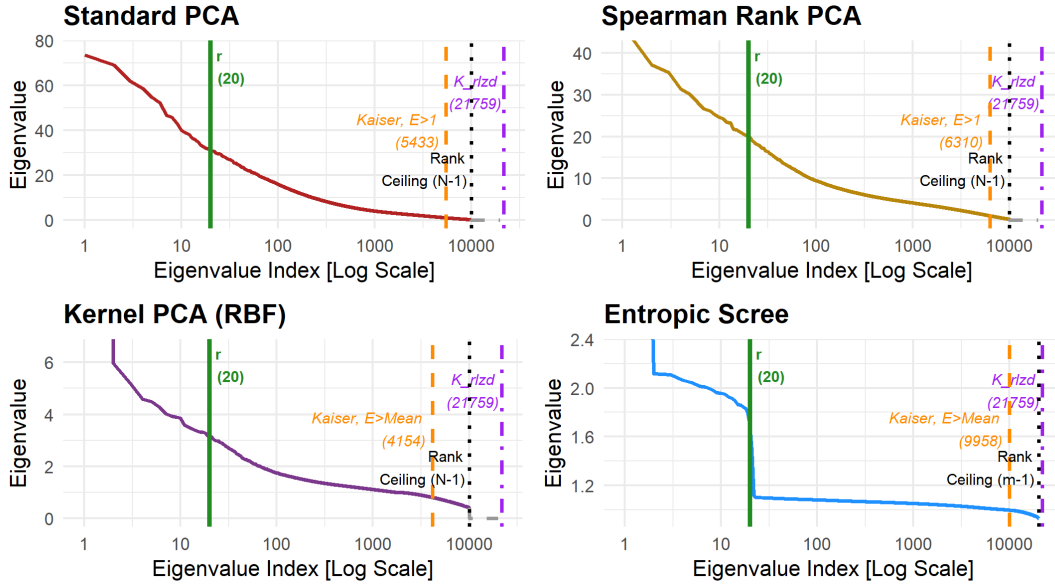
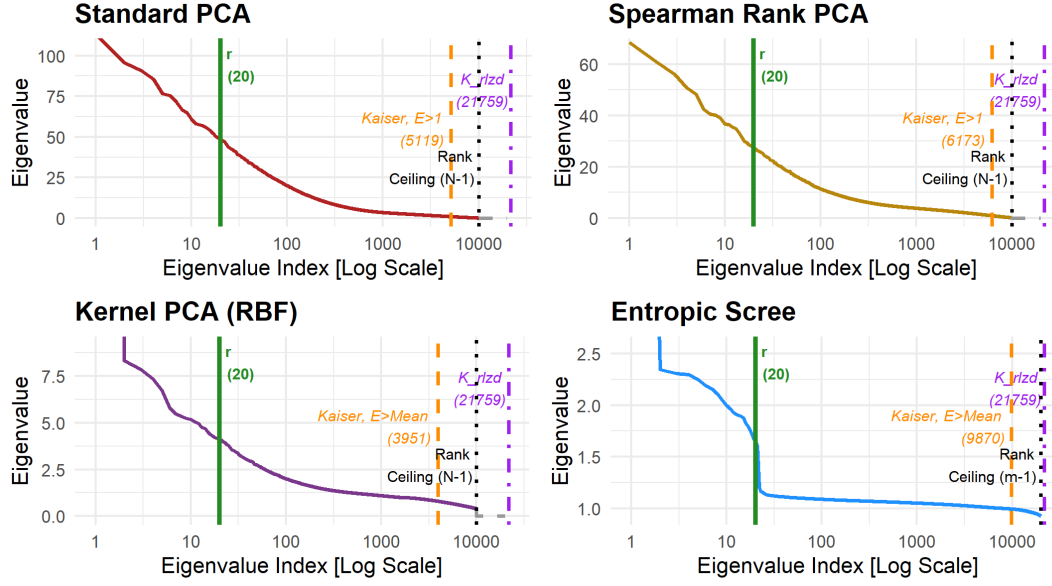
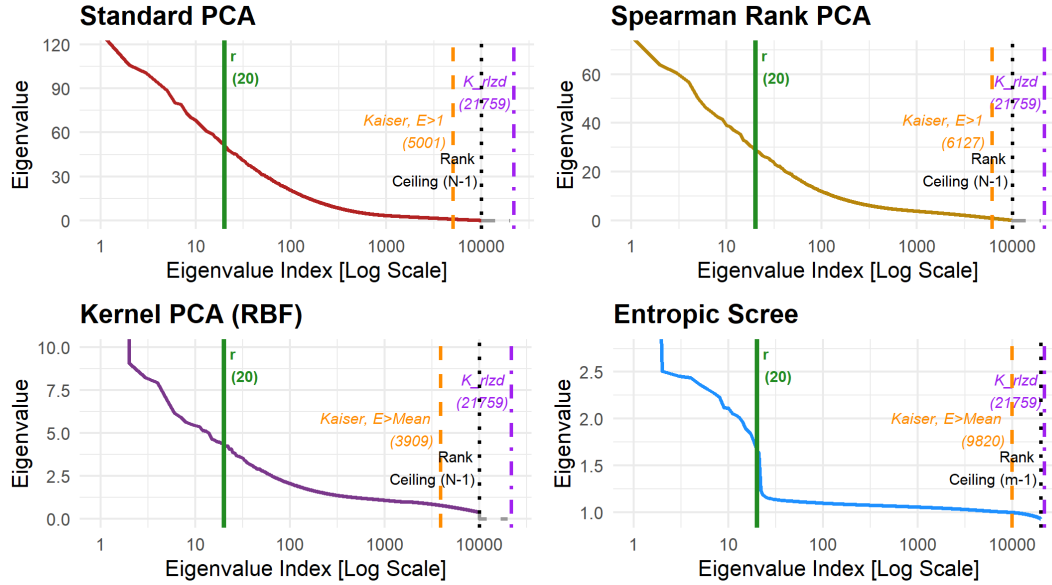


Figure 6: **Eigenspectra: Low Entanglement** ($\alpha_d = 15.0$).

The independent-Gaussian baseline was supplemented by runs with correlated Gaussian roots at the same stated sample size, measurement count and expansion complexity. The generator uses `clusterGeneration:rcorrmatrix`, a random partial-correlation construction, with $\alpha_d = \text{alphad}$ set to 15.0, 0.75, and 0.001 (Joe, 2006). Low, Medium and High identify these three generator settings.

In dimension 20, the marginal pair-correlation law is a symmetric beta distribution rescaled to $(-1, 1)$, with shape $a = \text{alphad} + 9$. Thus a very small `alphad` does not by itself imply that most root pairs have correlation near one. Matrix-level near dependence also requires examination of the joint spectrum.

The baseline comparison provides the starting point: standard PCA has no clearly resolved boundary at 20, while rank-PCA and the specified RBF kernel show a feature near 40 followed by an extended tail. These observations motivate examining what changes when


 Figure 7: Eigenspectra: Medium Entanglement ($\alpha_d = 0.75$).

 Figure 8: Eigenspectra: High Entanglement ($\alpha_d = 0.001$).

the generating coordinates become correlated. Section 8.3 discusses the possible relationship between the extracted directions and the generating transformations.

Across the reported correlated-root settings, the rank-PCA and RBF-kernel features near 40 were no longer comparably resolved. A larger early drop between indices 2 and 3 could dominate a secondary-gap rule, followed by an extended decline. One interpretation is that correlation concentrates weight into leading directions while smaller distinctions become less separated from the remainder.

Entropic Scree retained a transition near the generating count in these runs. In the highest setting, the selected primary boundary was $K_{roots} = 21$: axes 1 through 21 were retained. Relative to the nominal generating count of 20, this is a one-axis over-selection. The result indicates approximate stability under this setting.

The large measurement set offers many opportunities to observe different combinations, which is consistent with the measurement-breadth motivation. These runs do not isolate breadth as the cause of the retained boundary, because a separate breadth ablation was not reported. Distinct generating processes can remain hard to distinguish when they produce nearly identical measured consequences, even with many columns.

Perfectly dependent generating coordinates can become observationally redundant. More generally, distinguishing generating processes depends on the information retained by the available measurements and their discrete representations.

Appendix D. Interactions, Polynomial Folding and the Extended Signal Tail

The extended region motivates two questions: how can a low-dimensional generating system produce several informational directions, and which transformations preserve or discard measured distinctions? The following examples distinguish pairwise-visible from pairwise-invisible interactions and explain the effects of folding and monotone transformations.

Pairwise-Visible and Pairwise-Invisible Interactions

Consider independent standard-normal X_1, X_2 and $Y = X_1 X_2$. Then $\text{Cov}(X_1, Y) = 0$, yet

$$\text{Var}(Y \mid X_1 = x) = x^2.$$

The conditional distribution changes with $|x|$, so Y is not independent of X_1 . Pairwise information can therefore reveal an aspect of this interaction that covariance misses. In contrast, independent fair bits A, B with $Y = A \oplus B$ have zero MI for every pair despite joint determinism. An interaction is not necessarily pairwise visible or invisible merely because it involves several generating inputs.

When observed mixtures share some distinctions but differ in others, their pairwise information profiles can require multiple map directions. This motivates investigating whether repeatedly expressed interactions contribute to an extended spectral region. Purely pairwise-invisible structure can remain entirely absent from that region.

Folding and Loss of Resolution

For discrete $Y = f(X)$, $H(Y) \leq H(X)$ and $H(X, Y) = H(X)$, so

$$\mathcal{M}_{X,Y} = H(Y)/H(X)$$

when $H(X) > 0$. A strict entropy loss requires the transformation to merge states with positive probability. For example, let X be uniform on $\{-2, -1, 1, 2\}$ and $Y = X^2$. Then $H(X) = 2$ bits, $H(Y) = 1$ bit and $\mathcal{M}_{X,Y} = 1/2$. The squared value retains magnitude but loses sign. This is a concrete explanation of how a deterministic folded measurement can occupy a different information position from its source.

The measurement's relationships with all other variables in the matrix determine its spectral representation. For continuous variables, differential entropy does not obey the same transformation inequality, and a deterministic relationship can have infinite MI; the finite ratio used by this implementation is calculated after discretization.

Monotone Transformations and Equal-Frequency Binning

For a strictly increasing transform, such as X^3 , sample ordering is unchanged. Identical rank-based binning with consistent ties and missing-value handling therefore assigns the same categories to X and X^3 . Their empirical normalized MI is one when the binned entropy is nonzero. Different binning rules, inconsistent handling, numerical precision or additional measurement noise can change that correspondence.

What Can Be Bounded

If K_{rlzd} counts columns in a generating design, its centered linear rank is at most $\min(K_{rlzd}, N - 1)$. That fact does not transfer to a nonlinear pairwise MI matrix of the observed measurements. Exact centering gives $\text{rank}(\mathcal{M}_c) \leq m - 1$, and the exact positive-spectrum map has at most that many nonzero coordinate directions. The implemented boundary indices refer to the m working weights, including the numerical floor; their positions should therefore be distinguished from the number of strictly positive raw eigenvalues. A sharper relationship to generating-term counts requires additional assumptions and proof. The simulated boundary near $20 + \binom{20}{2}$ motivates further investigation of how generating-term structure relates to the extended spectrum.

References

- Charu C Aggarwal, Alexander Hinneburg, and Daniel A Keim. On the surprising behavior of distance metrics in high dimensional space. In *International Conference on Database Theory*, pages 420–434. Springer, 2001. URL https://doi.org/10.1007/3-540-44503-x_27.
- Simon Benjaminsson, Peter Fransson, and Anders Lansner. A novel model-free data analysis technique based on clustering in a mutual information space: Application to resting-state fMRI. *Frontiers in Systems Neuroscience*, 4:34, 2010. doi: 10.3389/fnsys.2010.00034.

- Kevin Beyer, Jonathan Goldstein, Raghu Ramakrishnan, and Uri Shaft. When is “nearest neighbor” meaningful? In *International Conference on Database Theory*, pages 217–235. Springer, 1999. URL https://doi.org/10.1007/3-540-49257-7_15.
- Kurt D. Bollacker and Joydeep Ghosh. Mutual information feature extractors for neural classifiers. In *Proceedings of International Conference on Neural Networks (ICNN’96)*, volume 3, pages 1528–1533. IEEE, 1996. doi: 10.1109/ICNN.1996.549127.
- Raymond B Cattell. The scree test for the number of factors. *Multivariate Behavioral Research*, 1(2):245–276, 1966. URL https://doi.org/10.1207/s15327906mbr0102_10.
- Claudio Ceruti, Simone Bassis, Alessandro Rozza, Gian Luca Lombardi, Elena Casiraghi, and Paola Campadelli. Danco: An intrinsic dimensionality estimator exploiting angle and norm concentration. *Pattern Recognition*, 47(8):2569–2581, 2014. URL <https://doi.org/10.1016/j.patcog.2014.02.013>.
- Sarvesh Chhetri. Mental health symptoms & illness prediction. Kaggle dataset, n.d. URL <https://www.kaggle.com/datasets/sarveshchhetri/mental-health-symptom-dataset-one-hot-encoded>. Accessed 25 September 2026.
- Jacob Cohen. The cost of dichotomization. *Applied psychological measurement*, 7(3):249–253, 1983. URL <https://doi.org/10.1177/014662168300700301>.
- Javier E. Contreras-Reyes. Mutual information matrix based on asymmetric Shannon entropy for nonlinear interactions of time series. *Nonlinear Dynamics*, 104(4):3913–3924, 2021. doi: 10.1007/s11071-021-06498-w.
- Thomas M. Cover and Joy A. Thomas. *Elements of Information Theory*. Wiley, 2nd edition, 2006. URL <https://doi.org/10.1002/0471200611>.
- Chandler Davis and William M Kahan. The rotation of eigenvectors by a perturbation. iii. *SIAM Journal on Numerical Analysis*, 7(1):1–46, 1970. URL <https://doi.org/10.1137/0707001>.
- Elena Facco, Maria d’Errico, Alex Rodriguez, and Alessandro Laio. Estimating the intrinsic dimension of datasets by a minimal neighborhood information. *Scientific Reports*, 7(1): 12140, 2017. URL <https://doi.org/10.1038/s41598-017-11873-y>.
- Marina Gomtsyan, Nikita Mokrov, Maxim Panov, and Yury Yanovich. Geometry-aware maximum likelihood estimation of intrinsic dimension. In *Proceedings of The Eleventh Asian Conference on Machine Learning*, volume 101, pages 1126–1141, 2019. URL <https://proceedings.mlr.press/v101/gomtsyan19a.html>.
- Nicholas J Higham. Computing a nearest symmetric positive semidefinite matrix. *Linear Algebra and its Applications*, 103:103–118, 1988. URL [https://doi.org/10.1016/0024-3795\(88\)90223-6](https://doi.org/10.1016/0024-3795(88)90223-6).

- Sune K. Jakobsen. Mutual information matrices are not always positive semidefinite. *IEEE Transactions on Information Theory*, 60(5):2694–2696, 2014. doi: 10.1109/TIT.2014.2311434.
- Qingchao Jiang and Xuefeng Yan. Nonlinear plant-wide process monitoring using MI-spectral clustering and Bayesian inference-based multiblock KPCA. *Journal of Process Control*, 32:38–50, 2015. doi: 10.1016/j.jprocont.2015.04.014.
- Harry Joe. Generating random correlation matrices based on partial correlations. *Journal of Multivariate Analysis*, 97(10):2177–2189, 2006. URL <https://doi.org/10.1016/j.jmva.2005.05.010>.
- Ian T. Jolliffe and Jorge Cadima. Principal component analysis: a review and recent developments. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 374(2065):20150202, 2016. doi: 10.1098/rsta.2015.0202. URL <https://doi.org/10.1098/rsta.2015.0202>.
- Terrence J. Lee-St. John, Jordan L. Lawson, and Bartłomiej Piechowski-Jozwiak. From garbage to gold: A data-architectural theory of predictive robustness. *arXiv preprint arXiv:2603.12288*, 2026. URL <https://arxiv.org/abs/2603.12288>.
- Elizaveta Levina and Peter J Bickel. Maximum likelihood estimation of intrinsic dimension. In *Advances in Neural Information Processing Systems*, volume 17, pages 777–784. MIT Press, 2004. URL https://proceedings.neurips.cc/paper_files/paper/2004/file/74934548253bcab8490ebd74afed7031-Paper.pdf.
- Feiya Lv, Shujian Yu, Chenglin Wen, and Jose C. Principe. Interpretable fault detection using projections of mutual information matrix. *Journal of the Franklin Institute*, 358(7):4028–4057, 2021. doi: 10.1016/j.jfranklin.2021.02.016.
- Jian Ma and Zengqi Sun. Mutual information is copula entropy. *Tsinghua Science and Technology*, 16(1):51–54, 2011. URL [https://doi.org/10.1016/S1007-0214\(11\)70008-6](https://doi.org/10.1016/S1007-0214(11)70008-6).
- Robert C MacCallum, Shaobo Zhang, Kristopher J Preacher, and Derek D Rucker. On the practice of dichotomization of quantitative variables. *Psychological methods*, 7(1):19–40, 2002. URL <https://doi.org/10.1037/1082-989X.7.1.19>.
- Qingfei Pan, Liang Ding, Siarhei Hladyszau, Xiangyu Yao, Jiayu Zhou, Lei Yan, Yogesh Dhungana, Hao Shi, Chenxi Qian, Xinran Dong, Chad Burdyschaw, Joao Pedro Veloso, Alireza Khatamian, Zhen Xie, Isabel Risch, Xu Yang, Jiyuan Yang, Xin Huang, Jason Fang, Anuj Jain, Arihant Jain, Michael Rusch, Michael Brewer, Junmin Peng, Koon-Kiu Yan, Hongbo Chi, and Jiyang Yu. scMINER: a mutual information-based framework for clustering and hidden driver inference from single-cell transcriptomics data. *Nature Communications*, 16:4305, 2025. doi: 10.1038/s41467-025-59620-6.
- Liam Paninski. Estimation of entropy and mutual information. *Neural Computation*, 15(6):1191–1253, 2003. doi: 10.1162/089976603321780272. URL <https://doi.org/10.1162/089976603321780272>.

- Robert Price. A useful theorem for nonlinear devices having gaussian inputs. *IRE Transactions on Information Theory*, 4(2):69–72, 1958. URL <https://doi.org/10.1109/TIT.1958.1057444>.
- Czesław Rajski. A metric space of discrete probability distributions. *Information and Control*, 4(4):371–377, 1961. URL [https://doi.org/10.1016/S0019-9958\(61\)80022-1](https://doi.org/10.1016/S0019-9958(61)80022-1).
- Olivier Roy and Martin Vetterli. The effective rank: A measure of effective dimensionality. In *15th European Signal Processing Conference*, pages 606–610. IEEE, 2007. URL <https://www.eurasip.org/Proceedings/Eusipco/Eusipco2007/Papers/a5p-h05.pdf>.
- Luis Gonzalo Sánchez Giraldo, Murali Rao, and Jose C. Principe. Measures of entropy from data using infinitely divisible kernels. *IEEE Transactions on Information Theory*, 61(1):535–548, 2015. doi: 10.1109/TIT.2014.2370058.
- Bernhard Schölkopf, Alexander Smola, and Klaus-Robert Müller. Nonlinear component analysis as a kernel eigenvalue problem. *Neural Computation*, 10(5):1299–1319, 1998. URL <https://doi.org/10.1162/089976698300017467>.
- Abe Sklar. Fonctions de répartition à n dimensions et leurs marges. *Publications de l’Institut de statistique de l’Université de Paris*, 8:229–231, 1959.
- Chudong Tong and Xuhua Shi. Mutual information based PCA algorithm with application in process monitoring. *CIESC Journal*, 66(10):4101–4106, 2015. doi: 10.11949/j.issn.0438-1157.20150374. In Chinese, with English abstract.
- Warren S Torgerson. Multidimensional scaling: I. theory and method. *Psychometrika*, 17(4):401–419, 1952. URL <https://doi.org/10.1007/BF02288916>.
- Ulrike von Luxburg. A tutorial on spectral clustering. *Statistics and Computing*, 17(4):395–416, 2007. doi: 10.1007/s11222-007-9033-z.
- Bo Wang and Yi Shen. A method on calculating high-dimensional mutual information and its application to registration of multiple ultrasound images. *Ultrasonics*, 44(Supplement 1):e79–e83, 2006. doi: 10.1016/j.ultras.2006.07.012.
- Xiaojun Zhao, Pengjian Shang, and Jingjing Huang. Mutual-information matrix analysis for nonlinear interactions of multivariate time series. *Nonlinear Dynamics*, 88(1):477–487, 2017. doi: 10.1007/s11071-016-3254-7.